Software Requirements Specification (SRS)

UTeM Micro-Credential Management System (MCMS)

Document Control

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Glossary

The terms below carry exactly the meaning fixed in the Glossary and Decision Log v0.6. Where this SRS adds a term that the Glossary does not define, it is marked *(SRS)*. Source tags such as [G 12.9] refer to clauses of the Garis Panduan Micro-Credential UTeM (the Guide).

Term	Definition
ABC	Academic Bank of Credit. Central digital depository of ACR. A deposit is storage and does not approve transfer. [G 12.5]
Academic authority	The body that takes final academic decisions such as approval to offer and stackable mapping: the Senate, whose decisions are recorded in MCMS by the deputy dean (D-11). Configured as a setting, not code.
ACR	Academic Credit Record. Academic record used in the ABC mechanism; prepared only for an eligible completed UMC or Quality-Verified SAMC. [G 12.4]
Achievement Record	UTeM internal record of modules passed, Digital Badges and MC completion (Rekod Pencapaian Pembelajaran). It is not an ACR. [G 8.7.5, 18.3]
Adapter	A defined interface to an external system (UCreds, ABC, QVC/MQA, payment, SSO, email) with a stub or manual version in the first release.
API	Application Programming Interface.
Audit trail	Append-only record of approvals, credential actions, result changes and corrections. History is never erased. [G 10.5.1, 18.10]
Available / Provisionally Redeemed / Redeemed	Credit status under the Single Redemption Principle: not yet used; used in an approved transfer to a programme still in progress; used toward a qualification already awarded. [G 12.9, App F9]
Body of Knowledge (BoK)	The knowledge domains a target qualification must cover; used in stackable degree mapping. [G 14.5.2]
CLO / PLO	Course Learning Outcome / Programme Learning Outcome.
CODL	Centre for Open and Distance Learning. System administrator and technical operator of MCMS and operator of UCreds; takes no academic decision (D-12).
Deputy dean (PTj)	The faculty deputy dean who owns micro-credential offerings in MCMS: endorses, verifies, confirms quality and records decisions (D-28).
Cohort (Offering)	One run of an approved MC version with its own dates, participants and results. *(SRS)*
Configurable value	A threshold or list kept in settings and never hard-coded (40 h SLT, 3-year QV limit, 10-year credit validity, 30% home credit, residency, alert days, retention).
CQI	Continuous Quality Improvement: finding, evidence, analysis, action, implementation and effectiveness check. [G 17]
Credit accumulation	Building a record of eligible credit over time. Separate from credit transfer. [G 12]
Credit transfer	Evaluation and acceptance of credit by the programme that receives it; in MCMS decided by the deputy dean of the faculty that owns the degree (D-11). Equivalence is judged on learning outcomes, not on credit or SLT alone. [G 13]
Credit validity	Credit in ABC is valid 10 years from deposit date. This differs from ACR retention (indefinite unless the owner or authority directs otherwise). [G 12.8]
Currency of knowledge	How up to date and relevant earlier learning still is when it is evaluated for equivalence or transfer. [G 4]

Term	Definition
Digital Badge	Proof of achievement of one module or unit. Not academic credit and not an ACR. [G 8.7.1, App E1]
Digital Credential / Certificate	Proof of completing the whole MC (Sijil / Digital Credential). Does not by itself give credit or transfer. [G 8.7.3, App E1]
HEPA	Student Affairs and Alumni Department. Assumed to register outside applicants together with Pejabat Pendaftar.
Material change	Change to outcomes, SLT, assessment, module structure, credit, MQF level, parent course link or QV status. Needs review and re-approval and is versioned. [G 9.5, 17.7, 17.8]
MC version	One approved, immutable definition of a micro-credential (outcomes, SLT, modules, assessment, credential). A material change creates a new version. *(SRS)*
MCE	Micro-Credential Eksekutif. Non-credit short learning under 40 hours SLT. Never carries credit unless reclassified and re-approved as a SAMC. [G 6.3.3]
Micro-Credential (MC)	Certification of a small, focused unit of learning with defined outcomes and assessment. Recognition follows achievement, not attendance. [G 6.1]
Module	A smaller unit of a UMC. Completion earns a Digital Badge, not credit and not whole-UMC completion. [G 8.7.1]
MQA / MQF	Malaysian Qualifications Agency / Malaysian Qualifications Framework.
OIDC	OpenID Connect, the sign-in protocol assumed for the Microsoft education account (OI-03). *(SRS)*
Participant	Any person enrolled or applying to enrol in an MC offering: UTeM student, staff or outside applicant (user class P8).
PDPA	Personal Data Protection Act 2010 (Act 709), with current amendments.
PTj	Pusat Tanggungjawab: the faculty, school, institute or centre that owns an offering. [G 15.2.2]
QA / QV / QVC	Quality Assurance applies to every MC. Quality Verification is the SAMC-specific check run by the QVC under MQA rules. Approval to offer is a separate decision from QV. [G 7.6, 11.2]
Quality-Verified SAMC	A SAMC that has passed QV. QV is valid for at most 3 continuous years and needs one completed cohort before submission. [G 11.2.1, 11.2.2]
RACI	Responsible, Accountable, Consulted, Informed: the role matrix of Guide Appendix H.
RBAC	Role-Based Access Control.
Release 1 / 2 / 3	Delivery stages of MCMS. Release 1 is the first buildable release (prototype scope). See Section 1.2 and decision D-24.
RPO / RTO	Recovery Point Objective / Recovery Time Objective. *(SRS)*
SAMC	Stand-alone Micro-Credential. Own offering, at least 40 hours SLT. Credit and MQF level exist only after QV. [G 6.3.2]
SDD	Software Design Description.
Single Redemption	Credit used for one qualification cannot be used for another. Transfer makes credit provisionally redeemed; completing the programme makes it redeemed. [G 12.9]
SIS	Student Information System of UTeM. Outside MCMS scope; a future adapter. *(SRS)*

Term	Definition
SLT	Student Learning Time: all learning and assessment time, not only contact time. [G 8.3]
SME	Subject-Matter Expert from industry or a profession (user class X2). [G 16.7]
SRS / PRD	Software Requirements Specification / Product Requirements Document.
SSO	Single Sign-On.
Stackable Degree	Planned pathway where eligible credit counts toward part of a degree. Needs admission, transfer approval, at least 30% credit earned at UTeM, residency (1 semester for conventional programmes), PLO mapping and graduation requirements. [G 14]
UCreds	UTeM online learning platform for micro-credentials; the pilot learning platform. Reached only through an adapter.
UMC	Unbundled Micro-Credential. Made by unbundling an accredited credit course; keeps its parent link; total SLT equals the parent course; whole UMC is at least 40 hours SLT or one credit. [G 6.3.1]
WCAG	Web Content Accessibility Guidelines (version 2.1 is the target).

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) gives a complete description of the UTeM Micro-Credential Management System (MCMS). MCMS is a proposed system that manages the whole life of a UTeM micro-credential: proposal and approval, offering and enrolment, learning results and credentials, Academic Credit Record (ACR) and Academic Bank of Credit (ABC) tracking, credit transfer, stackable degree pathways, and monitoring with Continuous Quality Improvement (CQI).

The document states **what** the system shall do, for whom, and under which rules. It does not state how the system is built; code structure, screen layouts and database scripts belong to the Software Design Description (SDD). The SRS is the agreed basis for the PRD, the SDD, development, and acceptance testing. Every requirement has a stable identifier so that it can be traced back to the Guide clause it comes from and forward to the design and the tests that prove it (decision D-01).

The document is written for two readers at once: a supervisor who reviews it, and developers and AI development agents who use it as the rule set for building the system (D-05). For that reason each requirement is a single statement, the source of every business rule is cited, and every assumption is visible.

1.2 Document Conventions

This document follows these conventions:

Term	Description
SHALL	Mandatory requirement that must be fulfilled in **Release 1**, the first buildable release (the Release 1 prototype of D-31, decision D-22). The developer is required to cover this feature in Release 1.
SHALL NOT	Prohibition that applies from Release 1 onward.
SHOULD	Requirement planned for **Release 2**. The developer should take it into account in the design so that it can be added without rework.
MAY	Requirement anticipated for **Release 3 or later**. The developer is encouraged to keep it in mind for future scalability.
TBD	To Be Determined. Information not yet available that will be provided in a later version. Every TBD is linked to an open item (OI-xx) in Appendix C.
Note	Provides additional information or clarification.

> **Note:** A requirement tagged with a Guide source, for example [G 12.9], states a rule that is mandatory for the complete system under UTeM policy. Its SHALL / SHOULD / MAY wording states only the release in which MCMS implements it. Guide rules are never relaxed by the release plan (D-24). Where a Guide rule is planned for a later release, Release 1 SHALL NOT contradict it; for example, it may block an action that it cannot yet support.

Requirements are categorised as follows:

Requirement Number	Description	Section
FR-XX	Functional Requirements	Section 3
BR-XX	Business Rules taken from the Guide (D-25)	Section 4
IR-XX	Interface Requirements	Section 5

Requirement Number	Description	Section
DR-XX	Data Requirements	Section 6
NFR-XX	Non-Functional Requirements	Section 7
SR-XX	Security Requirements	Section 7.2
D-XX / A-XX / OI-XX	Decision / Assumption / Open item, as kept in the Glossary and Decision Log	Appendix C

Each requirement table has three columns: **ID**, **Requirement** and **Source**. The Source column cites the Guide clause (`G 12.9`), Guide appendix table (`App E5`), decision (`D-17`) or assumption (`A-04`) that justifies the requirement. An empty source means the requirement is a good-practice or usability need introduced by this SRS. Identifiers are stable: once issued, an ID is never reused for a different requirement, and a withdrawn requirement keeps its ID with the word *Withdrawn*.

Configurable values are written in code font, for example `credit.abc_validity_years`. Each one is a setting that the System Administrator can change (D-16); the default value and its source are listed in Table 7.

Diagrams use common notations: flowcharts for processes, state diagrams for the life of a record, sequence diagrams for interactions between people and systems, and entity-relationship diagrams for the data model. Diagrams explain the requirements; where a diagram and a requirement differ, the requirement prevails.

1.3 Intended Audience

This document is intended for the following stakeholders:

Stakeholder	Role	Read first
Supervisor	Reviews the requirements and the decisions they rest on.	Sections 1, 2, 4, Appendix C
CODL / UTeM reviewers	Confirm assumptions and open items; assumed system owner.	Sections 1, 2, 3, Appendix C
Development team and AI development agents	Design and implement the system, using the requirement IDs and business rules as fixed rules.	Sections 2.3, 3, 4, 5, 6
Testers and quality assurance	Derive acceptance tests from requirement IDs and business rules.	Sections 3, 4, 7
Architects and security reviewers	Check the architecture, interfaces, data and security needs.	Sections 2, 5, 6, 7
System administrators	Maintain and configure the system after delivery.	Sections 3.1, 5, 7.7

1.4 Project Scope

UTeM offers micro-credentials in three categories: Unbundled Micro-Credentials (UMC) taken from accredited credit courses, Stand-alone Micro-Credentials (SAMC), and the non-credit Micro-Credential Eksekutif (MCE). The Garis Panduan Micro-Credential UTeM sets strict rules for each category. Most importantly, it separates approval, achievement, credentials, Quality Verification, ACR and ABC deposit, credit transfer and the award of a qualification into distinct steps that must never be merged [G 12–14, 18.14]. These rules are difficult to apply consistently with documents and spreadsheets. MCMS gives the University one system in which each step is a separate, traceable record.

MCMS is a new, standalone web system with its own database, sign-in handling and workflow (D-06). Learning content stays on the UCreds platform, and other external services are reached through adapters.

In Scope

The system will include the following key components:

- **Identity, roles and settings:** single sign-on with the Microsoft education account, role- and scope-based access, configurable thresholds and approving bodies.
- **Curriculum management:** category determination for UMC, SAMC and MCE, the proposal and design workspace of Guide Appendix A, review, approval, versioning and material-change control.
- **Offering and delivery:** readiness checks (Appendix B), public catalogue, registration, enrolment, cohorts, a dummy payment service, and import of learning results from UCreds.
- **Achievement and credentials:** assessment verification, the internal Achievement Record, Digital Badges, Certificates and Digital Credentials in an open signed format, and public verification without login.
- **Quality Verification for SAMC:** QV readiness evidence (Appendix D), QV status, validity and expiry control.
- **Credit recognition:** ACR preparation, participant consent, ABC deposit tracking, credit validity, credit transfer evaluation, the Single Redemption Principle and stackable degree pathway checks (Appendices F and G).
- **Governance:** monitoring, CQI (Appendix C), periodic review, suspension and termination, the append-only audit trail, records management and reports (Appendix H).
- **Bilingual operation:** screens, notifications and credentials in Bahasa Melayu and English (D-02).

Out of Scope

The following items are explicitly excluded:

1. Delivering learning content; this stays on UCreds or another learning platform.
2. The national ABC platform and the QVC/MQA procedures themselves. MCMS prepares and tracks; UTeM staff submit (D-14).
3. The full student information system, programme admission and degree programme regulations. MCMS records the results of these decisions but does not take them.
4. The UTeM payment hub itself; a dummy payment service stands in for Release 1 (A-03).
5. Open public self-registration; outside applicants first register at Pejabat Pendaftar or HEPA (D-10).
6. Partner-delivered micro-credentials run by other institutions (possible later).
7. Hardware procurement, and native mobile applications.

Release Overview

The release in which each capability is delivered follows from the SHALL / SHOULD / MAY wording of its requirements. Table 1 summarises the result. The PRD will hold the detailed release plan (D-22).

Table 1: *Summary of capabilities by release*

Capability area	Release 1 (SHALL)	Release 2 (SHOULD)	Release 3+ (MAY)
Identity and access	SSO (stub outside production), roles with scope, separation of duties, settings	Delegation of approval roles	–
Curriculum	Category rules, proposal workspace, review, approval, versioning, material change	Version comparison, batch meeting decisions, parent-course change alerts	–
Offering and enrolment	Readiness, catalogue, registration, dummy payment, cohorts	Waitlist, fee waivers, bulk enrolment	Live UTeM payment hub

Capability area	Release 1 (SHALL)	Release 2 (SHOULD)	Release 3+ (MAY)
Learning platform	CSV import and export with UCreds	Live UCreds connector	Near-real-time synchronisation
Credentials	Badges, certificates, signed Digital Credentials, public verification, correction and revocation	Signature check of uploaded credentials, sharing to professional networks	Credential wallet integration
Quality Verification	QV status, validity, expiry alerts, evidence checklist, manual submission package	–	Electronic submission to QVC/MQA
ACR, ABC and transfer	ACR, consent, manual ABC tracking, credit transfer, single redemption	External credit records, equivalence precedents	Live ABC connector
Stackable degree	Pathway record, 30% and residency checks, individual plan	PLO and Body of Knowledge matrices	SIS link for admission and graduation
Governance	Audit trail, corrections, traceability, standard reports, retention settings, cohort indicators, CQI records, periodic review, suspension and termination	Monitoring dashboards, facilitator and industry feedback, record holds	Analytics

1.5 References

This SRS is based on the following documents and sources:

1. Garis Panduan Micro-Credential UTeM (the Guide), CODL UTeM, Bab 1–18 with Lampiran A–H: the copy held in the project, Versi 1.0, 18 September 2026 (D-03).
2. Glossary and Decision Log for the UTeM MCMS, version 0.6, 27 October 2026 (MCMS_FND_V0.6_Oct_2026). This SRS is written against that version.
3. Software Requirements Specification, UTeM MCMS, informal draft v0.1, 23 October 2026 (input material only, D-04).
4. Software Requirements Specification for an AI-Driven Job Matching Platform, v1.0, May 2025. Used as a format reference only; no content is reused (D-26).
5. MQA: Guidelines to Good Practices: Micro-Credentials; Quality Verification of Stand-alone Micro-Credentials; Guidelines on the Academic Bank of Credit; Malaysian Qualifications Framework, second edition; Code of Practice for Programme Accreditation: Open and Distance Learning.
6. Guidelines on Developing Academic Programme through Stackable Degree Pathway.
7. UTeM academic regulations, academic quality assurance policy, and records and document management policy.
8. ISO/IEC/IEEE 29148, Systems and software engineering: Requirements engineering.
9. 1EdTech Open Badges 3.0 and W3C Verifiable Credentials Data Model (credential format guidance).
10. WCAG 2.1; OWASP Application Security Verification Standard 4.0; Personal Data Protection Act 2010 (Act 709) with current amendments.

2. Overall Description

2.1 Product Perspective

MCMS is a new system. It does not extend UCreds or the student information system. It has its own database, sign-in handling and workflow, and it runs on UTeM-controlled servers (D-06, constraint C1, C2). MCMS is the system of record for micro-credential proposals, approvals, versions, offerings, enrolments, verified achievements, credentials, ACR preparation and the University's own credit-use records. It is **not** the system of record for learning content (UCreds), national credit storage (ABC), Quality Verification decisions (QVC/MQA) or programme admission and graduation (student information system); it stores references to those decisions.

MCMS talks to every external system through an **adapter**: a defined interface with a stub or manual implementation in Release 1 that can later be replaced by a live connector without changing the business modules (D-06, D-14). The context diagram below shows who uses MCMS and which systems it connects to.

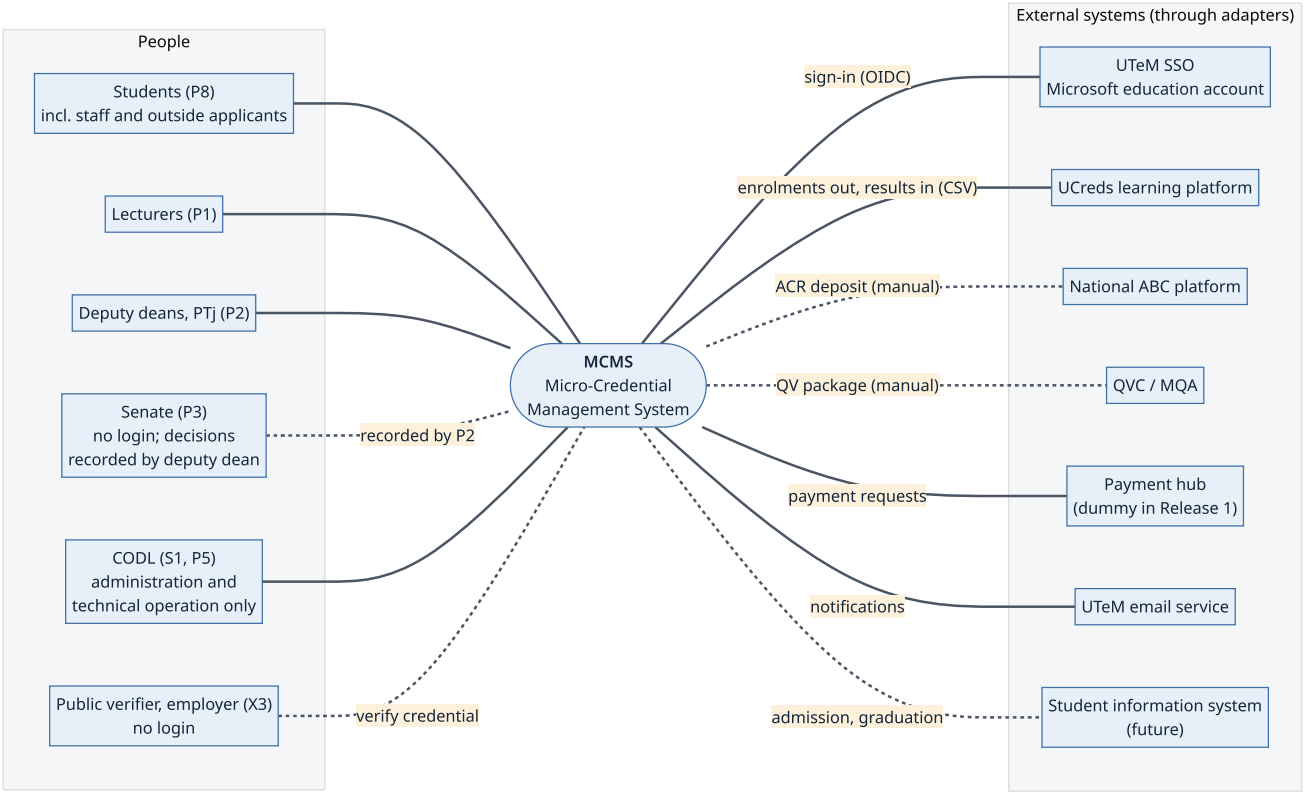


Figure 1: *System context of MCMS. Solid lines are automated exchanges in Release 1; dotted lines are manual exchanges or later integrations.*

2.2 Product Functions

MCMS will provide the following major functions. They are grouped into sixteen modules (M1–M16) carried from the draft SRS. The modules form a **modular monolith**: one deployable application with clearly separated modules that call each other through internal interfaces (D-07, proposed).

1. **Identity and governance foundation**: sign-in, roles and scopes, settings, approving bodies, notifications and the integration hub.
2. **Curriculum management**: category determination, proposal and design workspace, review and approval, versioning and material change.

3. **Offering and delivery**: readiness checks, catalogue, registration and enrolment, connection to the learning platform, assessment and verification of achievement.
4. **Recognition**: Achievement Record, Digital Badges, Certificates, Digital Credentials, public verification, and Quality Verification for SAMC.
5. **Credit**: ACR, ABC deposit tracking, credit validity, credit transfer, single redemption and stackable degree pathways.
6. **Quality and records**: monitoring, CQI, periodic review, suspension and termination, audit trail, records and reports.

Table 2: *Modules of MCMS*

Module	Function	Main Guide source	SRS section
M1	Identity, roles and access	Bab 15–16, 18.11, App H	3.1
M2	Category determination	6.3–6.5	3.2
M3	Proposal and design workspace	Bab 8–9, App A	3.2
M4	Review, approval and versioning	9.4–9.5, 15.4, 17.7–17.8	3.3
M5	Offering, catalogue and readiness	9.6, 10.3.1, App B	3.4
M6	Registration, enrolment and cohorts	10.3	3.5
M7	Learning platform connection	10.1–10.2	3.6
M8	Assessment and achievement verification	8.6, 10.4	3.7
M9	Achievement Record (internal)	8.7.5, 12.2, 18.3	3.8
M10	Credential issuance and verification	8.7, 10.5, App E	3.9
M11	Quality Verification management (SAMC)	11.2, App D	3.10
M12	ACR, ABC tracking and redemption	Bab 12, App F	3.11, 3.13
M13	Credit transfer and stackable pathway	Bab 13–14, App F–G	3.12, 3.14
M14	Monitoring, CQI and change control	Bab 11, 17, App C	3.15
M15	Records, audit and reporting	Bab 18, App H	3.16
M16	Notifications and integration hub	Bab 10, 18	3.17

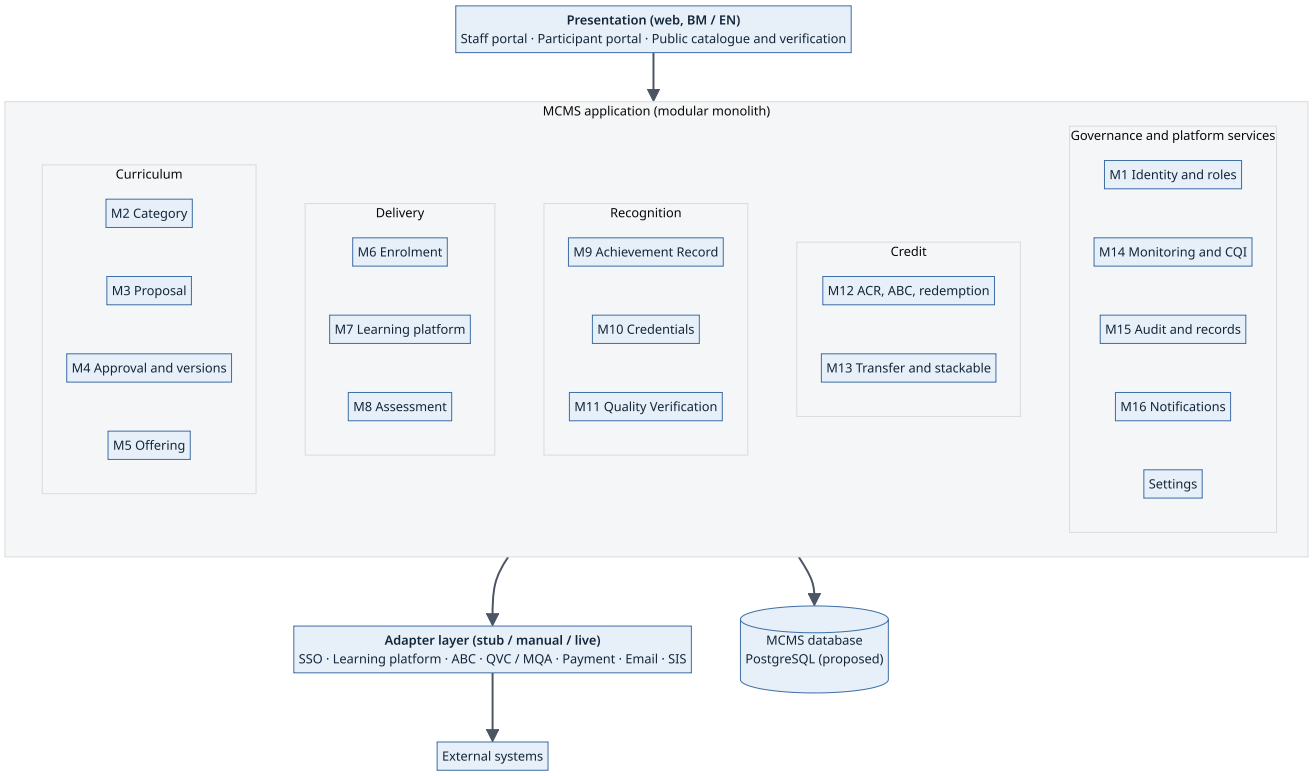


Figure 2: *Logical architecture: one application with separated modules, a single database, and an adapter for every external system.*

2.3 The Central Rule: Separate Recognition Stages

The Guide treats each of the stages below as a different thing with its own decision and its own record, and it warns repeatedly against merging them [G 12–14, 15.4, 18.14]. MCMS SHALL store them as separate records with separate decisions, so that none can be skipped or mistaken for another (D-15). This is the single most important rule for developers. It overrides any convenience shortcut: no single status field, screen action or button may combine two stages.

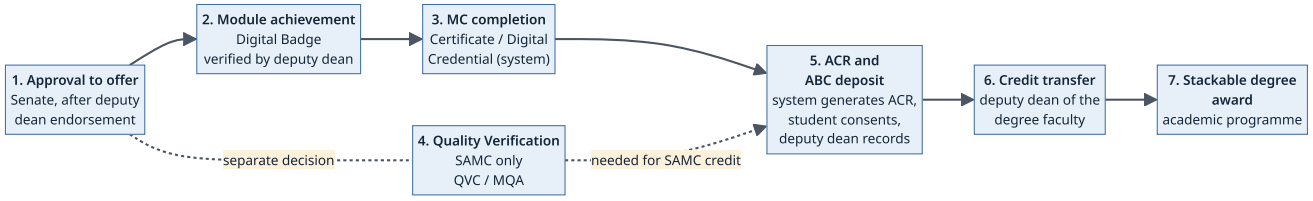


Figure 3: *The seven recognition stages. Each arrow is a separate decision by a separate party; reaching one stage never grants the next.*

Table 3: *Recognition stages, who decides, and what each stage does not imply*

Stage	Decided by	Record kept in MCMS	Does NOT imply
1. Approval to offer	Senate, after deputy dean endorsement; recorded by the deputy dean	Approval decision on an MC version	Credit, QV, transfer
2. Module achievement (Digital Badge)	Lecturer assesses; deputy dean verifies; system issues the badge	Module result, badge record	Whole-MC completion, credit

Stage	Decided by	Record kept in MCMS	Does NOT imply
3. MC completion (Certificate / Digital Credential)	Deputy dean confirms; system issues the credential	Achievement Record status, credential record	Credit, ACR, transfer
4. Quality Verification (SAMC only)	QVC / MQA	QV record with reference, credit, MQF level, validity	Transfer or admission
5. ACR and ABC deposit	System generates ACR; deputy dean confirms; student consents; deputy dean records the deposit	ACR, consent, deposit record	Transfer approval
6. Credit transfer	Deputy dean of the faculty that owns the degree	Transfer application, equivalence assessment, decision	Admission, degree award
7. Stackable degree award	UTeM academic programme	Reference to the award recorded by the programme	Nothing further

The life of a single micro-credential follows the cycle below [G 8, 15.3]. Each step has a responsible party, a review or approval mechanism and a record.

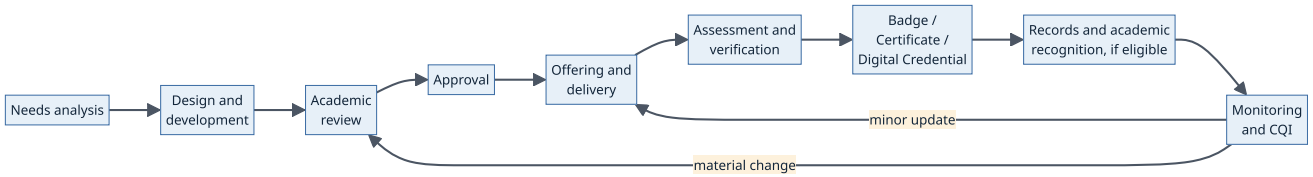


Figure 4: *Micro-credential life cycle. A minor update returns to delivery; a material change returns to review and re-approval.*

2.4 Micro-Credential Categories

Every rule in MCMS depends on the category of the micro-credential. The category is proposed early, checked at review and fixed by approval [G 6.5]. Table 4 gives the differences that the system enforces.

Table 4: *Comparison of UTeM micro-credential categories (Guide Table 1, clauses 6.3–6.5)*

Criterion	UMC	SAMC	MCE
Origin	Unbundled from a credit course of an accredited UTeM programme; keeps its parent link	Own offering, not a formal part of an accredited programme	Own non-credit short learning
SLT	Total equals the parent course; whole UMC at least 40 h or 1 credit	At least 40 h	Under 40 h
Credit	Possible, subject to academic rules; only the whole UMC carries credit	Only after Quality Verification	None
Quality route	Quality assurance of the parent programme	Internal QA, plus QV if credit or MQF level is sought	Internal UTeM QA
ABC deposit	Yes, if eligible (formal learning)	Only if Quality-Verified (non-formal learning)	No
Credit transfer	Yes, subject to equivalence	Only if Quality-Verified	No

Criterion	UMC	SAMC	MCE
Stackable degree	Yes, subject to conditions	Only if Quality-Verified	No
Credential	Badge per module; Certificate / Digital Credential for the whole UMC	Certificate / Digital Credential; credit and MQF only after QV	Certificate / Digital Credential stating non-credit completion

2.5 User Classes, Characteristics, and Needs

MCMS has six kinds of user (D-28). The codes keep the numbering of Guide Appendix H2 so that requirements stay traceable to the Guide; the Guide's other general roles (a separate quality unit P4, records and credential unit P6, ABC depositor P7 and receiving-programme evaluator X4) are not used as separate users, and their tasks are carried by the deputy dean or done automatically by the system (D-28, D-29). A person may hold several roles, each with its own scope (Section 3.1.2).

Table 5: *Users of MCMS*

User (code)	Who	Logs in?	Main functions in MCMS
Student (P8)	UTeM students and staff, graduates, professionals and members of the public registered by Pejabat Pendaftar or HEPA	Yes	Browse the catalogue; register and pay; follow module progress; view, download and share credentials; give or decline consent for ABC deposit; view the credit ledger; apply for credit transfer.
Lecturer (P1)	Academic staff of a PTj	Yes	Propose and design micro-credentials; teach on UCreds; enter and check results; perform the quality review and CQI after each cohort; collect QV evidence for a SAMC. When assigned, reviews a colleague's proposal as reviewer (X1).
Deputy dean (P2)	Deputy dean of the PTj that owns the offering	Yes	Assign reviewers; verify category and endorse; record Senate decisions; confirm readiness; verify results and confirm completion (which triggers automatic credentials); confirm quality reviews, CQI and QV evidence; correct, replace or revoke credentials; confirm ACR and record ABC deposits; approve module credit (D-18). The deputy dean of the faculty that owns a degree decides credit transfer cases into it.
Senate (P3)	UTeM Senate	No	Gives final approval to offer, approves material changes and stackable degree mappings. Its decisions are recorded in MCMS by the deputy dean with the meeting reference (D-11).
CODL (S1, P5)	CODL system administrators and technical staff	Yes, administration only	Manage users, roles, settings, templates and reference data; exchange data with UCreds; monitor adapters. CODL takes no academic decision and holds no academic records role (D-12).
Public verifier (X3)	Employers and members of the public	No	Check a credential by link or QR code; see only the minimum permitted information.

User (code)	Who	Logs in?	Main functions in MCMS
Industry expert (X2, optional)	External subject-matter expert invited to a proposal	Yes, limited	Comment on competencies and relevance only; no decisions [G 16.7].

In addition, **the system itself** acts automatically: it issues Digital Badges, Certificates and Digital Credentials once the deputy dean has verified results or confirmed completion and the pre-issuance checks pass; it generates ACR for eligible learning; and it sends reminders and expiry alerts (D-29).

The Guide's general RACI matrix (Appendix H3) is mapped onto these users below. MCMS uses it to route tasks and notifications; the actual role holders are settings (D-11, D-12).

Table 6: *RACI matrix of the micro-credential life cycle, mapped to MCMS users (from Guide Appendix H3)*

Activity	Lecturer P1	Deputy dean P2	Senate P3	CODL S1/P5	System	Student P8
Proposal and design of an MC	R	A	I	–	–	–
Category check UMC / SAMC / MCE	R	A	I	–	Suggests	–
Academic review	R (reviewer)	A	–	–	–	–
Approval to offer	C	R (endorses, records)	A	–	–	–
Platform and materials preparation	R	A	–	R (UCreds)	–	–
Registration and enrolment	I	A	–	C	R	R
Learning delivery and assessment	R	A	–	C	–	R
Verification of achievement	C	A/R	–	–	–	I
Issue Digital Badge / Certificate / Digital Credential	I	A (triggers)	–	–	R	I
Correction, replacement, revocation	C	A/R	–	–	–	I
Monitoring and CQI	R	A	–	–	Calculates	C
QV evidence and maintenance (SAMC)	R	A	–	–	Links evidence	–
ACR preparation for eligible credit	I	A (confirms)	–	–	R (generates)	I
Consent to ABC deposit	–	I	–	–	–	R
Recording the ABC deposit	–	A/R	–	–	–	I
Credit transfer evaluation and decision	C	A/R (degree faculty)	–	–	Checks rules	R (applies)
Stackable degree mapping	R/C	R	A	–	–	I
Material change	R	R (endorses)	A	–	Detects	I
Suspension / termination	C	R	A	–	–	I

2.6 Operating Environment

The platform will operate in the following environment:

1. **Technical environment**

- A web application accessible through current versions of Microsoft Edge, Google Chrome, Mozilla Firefox and Apple Safari (latest two major versions).
- Fully responsive design for phones and tablets; no native mobile application.
- Hosted on UTeM on-premises servers (virtual machines or containers), with separate development, test and production environments.

2. **Software environment**
   - Laravel (PHP) application framework, a currently supported release (D-08).
   - PostgreSQL as the primary database; MySQL or MariaDB as an accepted fallback if CODL already hosts it (D-09, OI-02). Queries stay portable through Laravel's database layer.
   - A queue worker for background jobs (imports, credential issuance, notifications, alerts) and a scheduler for timed jobs (expiry alerts, retention checks).
   - UTeM single sign-on based on the Microsoft education account (OpenID Connect assumed, OI-03); UTeM SMTP email service.
3. **User environment**
   - Staff use office desktops and laptops on the campus network or remotely over the internet.
   - Participants use personal devices, often phones, with varying connection quality.
   - Verifiers open a credential link or scan a QR code without signing in.
   - Interface in Bahasa Melayu and English.
4. **Organisational environment**
   - Offices and data owners: PTj (academic owner of offerings and records), CODL (system operation), Pejabat Pendaftar, HEPA and UTeM ICT.

2.7 Design and Implementation Constraints

The following constraints will affect the design and implementation of the platform:

1. **Technical constraints**
   - C1: MCMS SHALL be a new standalone system, not an extension of UCreds or the student information system (D-06).
   - C3: Every external system SHALL be reached through an adapter with a stub or manual implementation for Release 1 (D-06, D-14).
   - Architecture: a modular monolith built on Laravel (D-07, D-08).
   - C8: Credentials SHALL use an open, signed format; blockchain is not required (D-19).
2. **Data and rule constraints**
   - C2: All data SHALL stay on UTeM-controlled infrastructure.
   - C4: Thresholds and lists SHALL be settings, not code (D-16).
   - C5: Credit value, MQF level and QV wording SHALL be generated only from verified records, never typed by hand (D-17).
   - C6: Approvals, credential actions, result changes and corrections SHALL be written to an append-only audit log (D-20).
   - The recognition stages SHALL be separate records (D-15).
3. **Regulatory constraints**
   - Compliance with the Guide, MQA guidelines and the MQF.
   - Compliance with PDPA 2010 and UTeM policies on records, information security and privacy.
4. **Business constraints**
   - Built by an intern as a concept and prototype; Release 1 is the prototype scope (D-22).
   - Maintainable by CODL staff who mainly use PHP (D-08).

- Approving bodies and role holders may change; the system must adapt by configuration (D-11, D-12).

5. **User constraints**
   - C7: Interface and credentials SHALL support Bahasa Melayu and English (D-02).
   - Accessible to users with disabilities (WCAG 2.1 AA) and usable on low-end phones.

2.8 User Documentation

The following user documentation will be developed with the system:

1. **Online help**: context help on each screen and a Frequently Asked Questions page that explains, in plain words, the difference between a badge, a certificate, an ACR, an ABC deposit and a credit transfer.
2. **User guides**, in Bahasa Melayu and English: Lecturer and Developer Guide; PTj and Approver Guide; Quality Assurance Guide; Records, Credential and ABC Guide; Participant Guide; System Administrator Guide.
3. **Technical documentation**: adapter interface guide, CSV import and export templates, credential format and verification guide, installation and operations guide.
4. **Training materials**: quick reference cards per role and short demonstration videos.

2.9 Assumptions and Dependencies

Assumptions

The assumptions below are carried from the Glossary and Decision Log. Each one is built so that, if it proves wrong, only configuration or a single adapter changes.

ID	Assumption	Effect if wrong
A-01	Superseded by D-11: deputy dean endorses, Senate approves (recorded by the deputy dean), degree-owning deputy dean decides transfers.	Approver bodies remain settings.
A-02	ABC and QVC/MQA accept submissions manually in Release 1.	A live adapter is added later.
A-03	UTeM's payment hub is not ready; a dummy payment service is used.	Replace one adapter.
A-04	UCreds is the pilot learning platform; its connection method is unknown.	CSV import is the fallback; connector design shifts.
A-05	UCreds already issues module badges and has its own database.	MCMS issues badges itself.
A-06	Outside applicants register at Pejabat Pendaftar or HEPA, then sign in with a Microsoft education account.	Add an applicant pre-registration screen.
A-07	Superseded by D-12 and D-29: CODL is administrator and technical operator only; credentials and ACR are produced by the system after deputy dean verification.	Role holders change in settings.
A-08	Retention periods are placeholders.	Edit settings.
A-09	Sizing is indicative: 200 offerings and 20,000 participants a year, peaks of 2,000 registering within hours.	Capacity plan changes.

Dependencies

1. Access to the UTeM SSO tenant and agreement on the sign-in protocol (OI-03).

2. Agreement with the UCreds team on the data exchange format (OI-04).
3. Availability of a database server and hosting environment at UTeM (OI-02).
4. Reference data for PTj, programmes and parent courses, including accreditation status, from the relevant offices.
5. Decisions on the approving bodies, retention periods and ownership of credential signing keys (OI-06, OI-07, OI-08).

3. System Features and Functional Requirements

This section details the functional requirements of MCMS, organised by system feature in the order of the micro-credential life cycle. Each subsection begins with a short description of the feature and, where it helps, a diagram, followed by a requirement table. Each requirement has a unique ID (FR-XX) for traceability. The business rules that these requirements enforce are listed with their own IDs in Section 4.

3.1 Identity, Roles and Access (M1)

Staff and participants sign in with their Microsoft education account through UTeM single sign-on; MCMS keeps no passwords of its own in production (D-10). Access is controlled by role and by scope: a person sees and acts only on the PTj and micro-credentials assigned to them. Approving bodies and role holders are settings, so a change in the University's committees does not need a code change (D-11, D-12).

3.1.1 Sign-in and Account Recognition

ID	Requirement	Source
FR-01	The system **SHALL** authenticate staff and participants through UTeM single sign-on using their Microsoft education account. The protocol is assumed to be OpenID Connect (TBD, OI-03).	D-10
FR-02	The system **SHALL NOT** offer open public self-registration. A local account **SHALL** be created only on a person's first successful SSO sign-in or by an import of applicants registered by Pejabat Pendaftar or HEPA.	D-10, A-06
FR-03	On first sign-in the system **SHALL** create a local user profile from the SSO claims (full name, email, staff or matric number where present) and **SHALL** assign the Participant role by default.	D-10
FR-04	The system **SHALL** recognise outside applicants registered by Pejabat Pendaftar or HEPA through a configurable account attribute or an imported applicant list (TBD, OI-03).	A-06
FR-05	In development and test environments the system **SHALL** provide a stub identity provider that lets testers sign in as any configured test user and role without the real SSO tenant. The stub **SHALL** be impossible to enable in production.	D-14
FR-06	The system **SHALL** end an idle session after `auth.idle_timeout_minutes` and **SHALL** provide a sign-out action that ends the local session.	
FR-07	A user **SHALL** be able to view their profile and edit only non-authoritative fields (preferred language, notification preferences, contact phone). Name and identifiers **SHALL** come only from the authoritative source.	G 18.11
FR-08	The sign-in screen **SHALL** provide explicit institutional format guidance for UTeM Office 365 credentials, distinguishing staff (`UserId@utem.edu.my`) and student (`MatricNo@student.utem.edu.my`) account patterns, with vertically centered presentation and test persona access isolated to non-production environments.	D-44
FR-09	Each user **SHALL** be able to choose Bahasa Melayu or English as interface language at any time; the choice **SHALL** be remembered across sessions.	D-02
FR-10	The system **SHOULD** let the System Administrator merge two local accounts that belong to the same person, keeping every linked record and writing the merge to the audit trail.	

3.1.2 Roles, Scopes and Separation of Duties

ID	Requirement	Source
FR-11	The system **SHALL** control access by role-based access control using the users of Section 2.5 (P1, P2, P3, P5, P8, X1–X3, S1). Permissions **SHALL** be checked on the server for every action, not only hidden in the interface.	G 15–16, App H
FR-12	A user **SHALL** be able to hold more than one role at the same time, each with its own scope.	App H2
FR-13	Every staff role assignment **SHALL** carry a scope: the whole University, one or more PTj, or one or more specific micro-credentials. A user **SHALL** see and act only on records inside their scope.	G 15.2.2
FR-14	Only the System Administrator (S1) **SHALL** assign or revoke roles. Each change **SHALL** record the role, scope, reason, effective date and the person making the change in the audit trail.	D-20
FR-15	The system **SHALL** prevent the same person from both submitting and deciding the same item. A lecturer **SHALL NOT** review or approve their own proposal, and the person who enters a result **SHALL NOT** be its only verifier.	G 15.1
FR-16	The System Administrator role **SHALL NOT** include permission to approve academic matters, verify results, issue credentials, prepare ACR or decide credit transfer.	G 15.2.3
FR-17	The approving body for each decision type (approval to offer, material change, transfer rules, individual transfer case, stackable mapping, suspension) **SHALL** be a setting that maps the decision type to a role. Changing it **SHALL NOT** need a code change.	D-11, A-01, OI-06
FR-18	A reviewer (X1) **SHALL** declare whether they have a conflict of interest before opening a review. A declared conflict **SHALL** remove the reviewer from that item and notify the person who assigned them.	G 16.6
FR-19	An industry expert or SME (X2) **SHALL** be able to read the proposal and add comments on competencies and relevance only; X2 **SHALL** have no decision rights.	G 16.7
FR-20	An approver **SHOULD** be able to delegate their approval role to another eligible user for a set period. The delegation **SHALL** be shown on every decision taken under it.	

3.1.3 System Administration and Settings

Thresholds from the Guide may be revised by UTeM or MQA, so they are kept as settings (D-16). Table 7 lists the settings that Release 1 must provide, with their default values.

Table 7: *Configurable settings and default values*

Setting key	Meaning	Default	Source
`slt.min_hours_samc`	Minimum SLT for a SAMC	40 hours	G 6.3.2, BR-01
`slt.min_hours_umc_whole`	Minimum SLT for a whole UMC (or 1 credit)	40 hours	G 6.3.1, BR-01
`slt.min_hours_module_credit_model`	Minimum SLT per module under the credit-based unbundling model	40 hours	G 6.3.1.1
`credit.slt_hours_per_credit`	Notional learning hours per credit	40 hours	MQF, D-27
`qv.max_validity_years`	Maximum continuous validity of QV	3 years	G 11.2.2, BR-02
`qv.min_completed_cohorts`	Completed cohorts needed before QV submission	1	G 11.2.1, BR-03
`credit.abc_validity_years`	Validity of credit in ABC from deposit date	10 years	G 12.8, BR-04

Setting key	Meaning	Default	Source
`stackable.min_home_credit_percent`	Minimum share of graduating credit earned at UTeM	30%	G 14.5.1, BR-05
`stackable.residency_semesters`	Residency for conventional programmes	1 semester	G 14.5.3, BR-06
`alerts.qv_expiry_days`	Days before QV expiry at which alerts are sent	180, 90, 30	D-16
`alerts.credit_expiry_days`	Days before credit expiry at which the participant is alerted	365, 90	D-16
`alerts.decision_overdue_days`	Days after which a pending review or decision is reminded	14	
`umc.completion_window_months`	Default period to complete all modules of a modular UMC; an offering may set its own period	TBD (no value until OI-13 is answered)	G 7.4 (OI-13)
`auth.idle_timeout_minutes`	Idle session timeout	30 minutes	
`retention.*`	Retention period per record category. The Release 1 prototype sets 2 years for the notification log, the report run log and the integration call log (`retention.notifications_years`, `retention.report_runs_years`, `retention.integration_years`); academic records stay TBD	TBD	G 18.12 (A-08, OI-07)
`verification.public_fields`	Fields shown on the public verification page	Name, credential, issuer, date, status	App E6
`approval.route.*`	Approving role per decision type: endorsement, senate, transfer, stackable, suspension, disposal	See 3.1.2; deputy dean in Release 1	D-11
`qv.provider_registered`	Whether UTeM is registered and authorised to offer the programme, checked before a SAMC can be ready to submit for QV	1 (yes)	FR-167
`quality.target_participation_percent`, `quality.target_completion_percent`, `quality.target_pass_percent`, `quality.target_issuance_percent`	Targets that the cohort indicators of Appendix C2 are compared with	80%, 70%, 70%, 95%	App C2
`review.interval_months`	Months between the periodic reviews of a micro-credential	24 months	G 17.9, FR-252
`review.remind_days`	Days before a periodic review at which the lecturer and PTj are reminded	30 days	FR-252
`demo.clock_offset_days`	Days the demo clock is ahead of the real date; exists only in a test environment and is refused in production	0	PRD 7.2, FR-05

ID	Requirement	Source
FR-21	The system **SHALL** keep every value of Table 7 in a settings store that S1 can edit through the interface. No business rule **SHALL** read a threshold from program code.	D-16

ID	Requirement	Source
FR-22	Each settings change **SHALL** record the old value, new value, editor, reason and effective date. A decision **SHALL** be evaluated with the value that was in force on the date of the decision.	D-16, D-20
FR-23	S1 **SHALL** be able to manage, in both languages, the templates for notifications, certificates, Digital Credential display and recognition statements. Template changes **SHALL** be versioned.	D-02
FR-24	S1 **SHALL** be able to search, view, deactivate and reactivate user accounts. Deactivation **SHALL NOT** delete or alter any record linked to the user.	G 18.1
FR-25	S1 **SHALL** maintain reference data for PTj, academic programmes (code, name, MQF level, mode, accreditation status, total graduating credits) and parent courses (code, name, credit, SLT, CLOs), by form or by CSV import.	G 6.3.1, App G1
FR-26	S1 and P5 **SHALL** see the mode of each adapter (stub, manual or live), the time of its last run and its recent errors.	D-14

3.2 Category Determination and Proposal (M2, M3)

A micro-credential starts as a proposal prepared by a lecturer in a guided workspace that follows Guide Appendix A. The first step is to determine the category, because the category decides the design rules, the approval route, the quality route and the possible recognition [G 6.5]. The decision logic is shown in Figure 5.

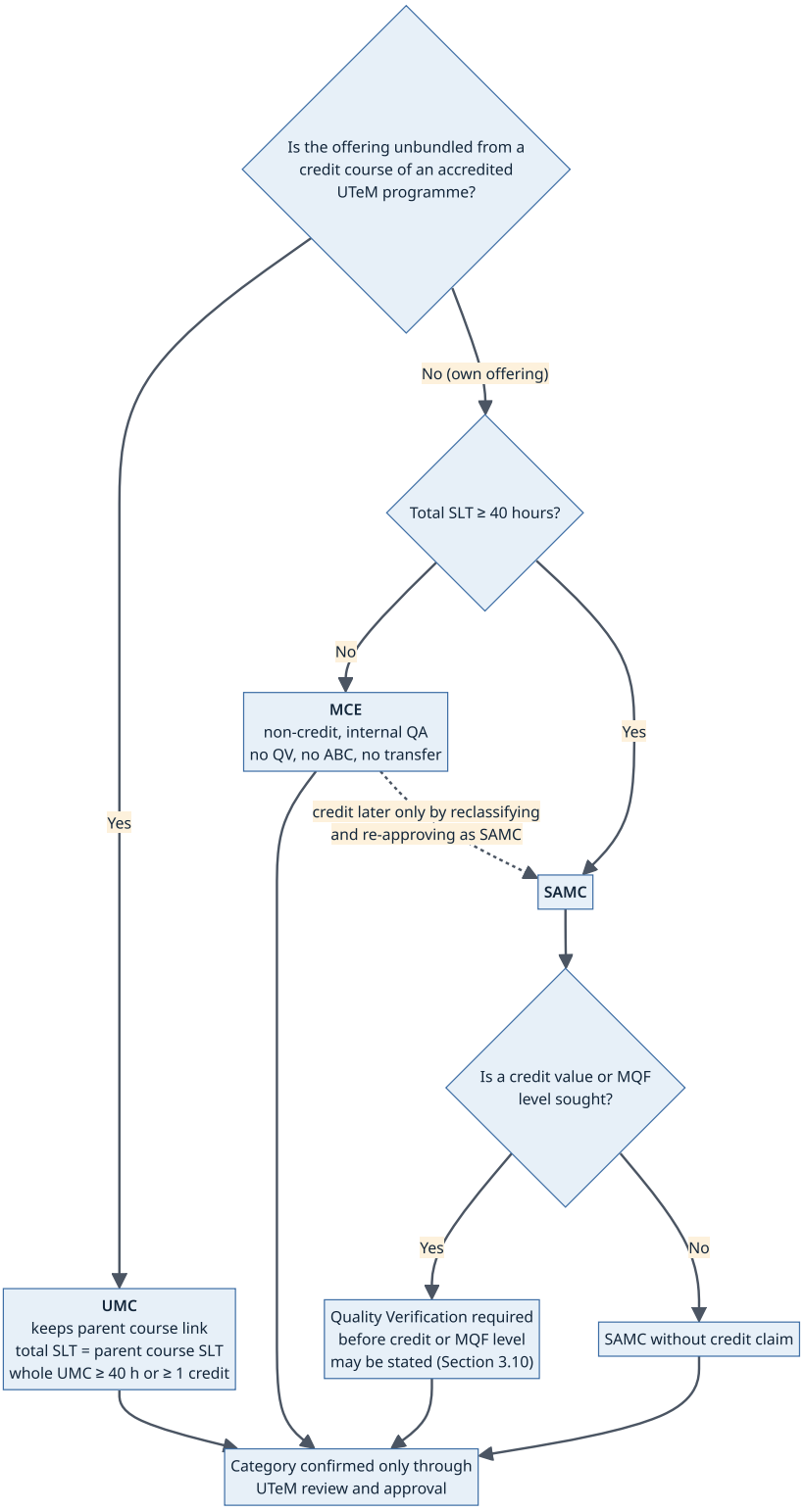


Figure 5: *Category determination, following Guide clause 6.5 and Figure 3 of the Guide.*

3.2.1 Category Determination

ID	Requirement	Source
FR-27	The system **SHALL** guide the proposer through category determination by asking for (a) the origin of the offering (unbundled from a credit course of an accredited UTeM programme, or an own offering), (b) the planned total SLT, and (c) whether a credit value or MQF level is sought.	G 6.5

ID	Requirement	Source
FR-28	From the answers the system **SHALL** propose a category according to Figure 5 and **SHALL** show the reason in plain words.	G 6.5
FR-29	When the origin is a credit course of an accredited UTeM programme, the system **SHALL** allow only the category UMC.	G 6.5
FR-30	When the offering is an own offering with total SLT below `slt.min_hours_samc`, the system **SHALL** allow only the category MCE and **SHALL** mark it non-credit.	G 6.3.3
FR-31	When the offering is an own offering with total SLT of at least `slt.min_hours_samc`, the system **SHALL** allow the category SAMC. If a credit value or MQF level is sought, the system **SHALL** record that QV is planned.	G 6.3.2
FR-32	The category **SHALL** stay "proposed" until it is verified by the PTj and fixed by approval. Only a confirmed category **SHALL** drive offering, credential and credit rules.	G 6.5, App H3
FR-33	The system **SHALL NOT** allow a module of a UMC to be proposed as a SAMC only because it can be completed separately.	G 6.2
FR-34	An MCE **SHALL** become eligible for credit only through a new proposal version that reclassifies it as a SAMC, followed by approval and QV. The system **SHALL** offer this as a "Reclassify" action that starts such a version.	G 6.3.3

3.2.2 Proposal and Design Workspace

The workspace collects the content of Guide Appendix A (template sections A1–A10). Fields that do not apply are marked "Not applicable (TB)" rather than left empty.

ID	Requirement	Source
FR-35	A lecturer (P1) **SHALL** be able to create a proposal of type New, Revision or Re-offer for a PTj inside their scope.	App A1
FR-36	The proposal **SHALL** capture the basic information of Appendix A1: • title in Bahasa Melayu and English; • owning PTj and lecturer (academic owner); • category (from 3.2.1) and proposal type; • field or discipline and target group; • delivery mode (face-to-face, online, blended, other) and delivery language; • proposed offering period.	App A1
FR-37	The proposal **SHALL** capture the justification of Appendix A2: the need or problem addressed, the evidence of need (industry or employer analysis, participant feedback, programme need, professional standard, technology change, other) with attachments, and the added value compared with existing courses or training.	G 9.1, App A2
FR-38	The proposer **SHALL** enter learning outcomes as separate items. Each outcome **SHALL** be linked to content, learning activities, assessment and SLT, as in the learning design table of Appendix A5.	G 8.2, App A5
FR-39	The system **SHALL** calculate total SLT from the design rows, show the split between learning activities and assessment, and show the result against the category threshold.	G 8.3
FR-40	The proposal **SHALL** capture the assessment design of Appendix A6: methods, weightings, pass or completion standard, identity and integrity mechanism, and feedback to participants. The system **SHALL** check that weightings total 100%.	G 8.6, App A6
FR-41	The system **SHALL** show an alignment matrix of learning outcomes against assessment components and **SHALL** warn about any outcome without assessment and any assessment component not linked to an outcome.	G 8.2, 9.3

ID	Requirement	Source
FR-42	The proposal **SHALL** capture the credential design of Appendix A7: credential types (Digital Badge, Certificate, Digital Credential, other), achievement level (module or whole MC), QV status for a SAMC (not applied, will apply, Quality-Verified, not applicable) and the verification method.	App A7
FR-43	Credit value and MQF level **SHALL** be entered only as "proposed" values. They **SHALL NOT** appear in the catalogue or on any credential until confirmed by approval (UMC) or by a QV decision (SAMC).	G 10.5, App D2, D-17
FR-44	The proposal **SHALL** capture the readiness items of Appendix A8 (materials, platform, assessment and rubrics, facilitators, participant information, records and credential mechanism, accessibility) with status Yes / Not yet / Not applicable and evidence notes.	App A8
FR-45	The proposer **SHALL** be able to attach supporting documents (PDF, Office documents, images) within a configurable size limit.	
FR-46	The proposer **SHALL** be able to add co-developers from the same PTj, who may edit the draft. Every edit **SHALL** be attributed to its author.	G 16.4
FR-47	The system **SHALL** save drafts automatically and **SHALL** let the proposer leave and resume a proposal.	
FR-48	The proposer **SHALL** confirm that third-party materials respect copyright and intellectual property rules, with a note on any licensed material.	G 9.3
FR-49	Before submission the system **SHALL** check the proposal against the category rules of Section 4 and **SHALL** list blocking errors (which prevent submission) and warnings (which need a written response).	G 9.4
FR-50	The system **SHALL** produce a printable proposal document in the layout of Appendix A for meetings and records.	App A

3.2.3 UMC Unbundling and Module Structure

A UMC must keep a clear academic link with its parent course: learning outcome mapping, SLT, content and assessment [G 6.3.1]. The unbundling model decides how modules and assessment are arranged [G 6.3.1.1].

ID	Requirement	Source
FR-51	For a UMC the system **SHALL** record the parent programme, the parent course (code, name, credit, total SLT, CLOs), the accreditation status of the programme, the unbundling model and the number of modules. These **SHALL** be selected from reference data, not typed.	App A4
FR-52	The system **SHALL** block submission of a UMC whose parent programme is not recorded as meeting the required accreditation status.	G 6.3.1, BR-09
FR-53	The proposer **SHALL** choose one unbundling model: credit-based (each module at least `slt.min_hours_module_credit_model` with its own learning, activities and assessment), topic-based (modules by topic, SLT may differ), or separate learning and assessment (learning modules plus a separate integrated assessment module).	G 6.3.1.1
FR-54	The system **SHALL** block submission unless the total SLT of all modules equals the SLT of the parent course.	G 6.3.1, BR-08
FR-55	The system **SHALL** block submission of a UMC whose whole structure has less than `slt.min_hours_umc_whole` or less than one credit.	G 6.3.1, BR-01
FR-56	Under the credit-based model the system **SHALL** block submission if any module has less than `slt.min_hours_module_credit_model`.	G 6.3.1.1(a)

ID	Requirement	Source
FR-57	Under the topic-based model the system **SHALL** permit individual module SLT to vary without requiring the 40-hour threshold, **SHALL** require at least two modules, and **SHALL** enforce that total module SLT strictly matches parent course SLT.	G 6.3.1.1(b), Rajah 6
FR-58	Under the separate learning and assessment model the system **SHALL** require at least one assessment module and **SHALL** check that the assessment structure as a whole covers every CLO of the parent course.	G 6.3.1.1(c), 8.6
FR-59	Each module learning outcome **SHALL** be mapped to one or more parent course CLOs. The system **SHALL** warn if any parent CLO is not covered by at least one module.	G 8.2, 9.4
FR-60	Each module **SHALL** have a flag "separately recognisable credit unit" with default No. The system **SHALL** let the flag be set to Yes only by a recorded case-by-case decision of the role configured as the academic authority for this decision type; the System Administrator role **SHALL NOT** be able to set it. The whole UMC always carries the credit of its parent course.	G 13.5, 14.2.1, 15.2.3, D-18
FR-61	When the reference record of a parent course changes in learning outcomes, content, SLT, assessment or credit, the system **SHOULD** flag every linked UMC for review and notify its lecturer and PTj.	G 11.5, 14.8

3.3 Review, Approval and Versioning (M4)

Every micro-credential must be reviewed and approved by the authority set by UTeM before it is offered [G 9.4–9.5]. Approval applies to one exact version of the structure. A material change creates a new version that needs review and re-approval before it takes effect [G 17.7–17.8]. Figure 6 shows the life of a proposal version.

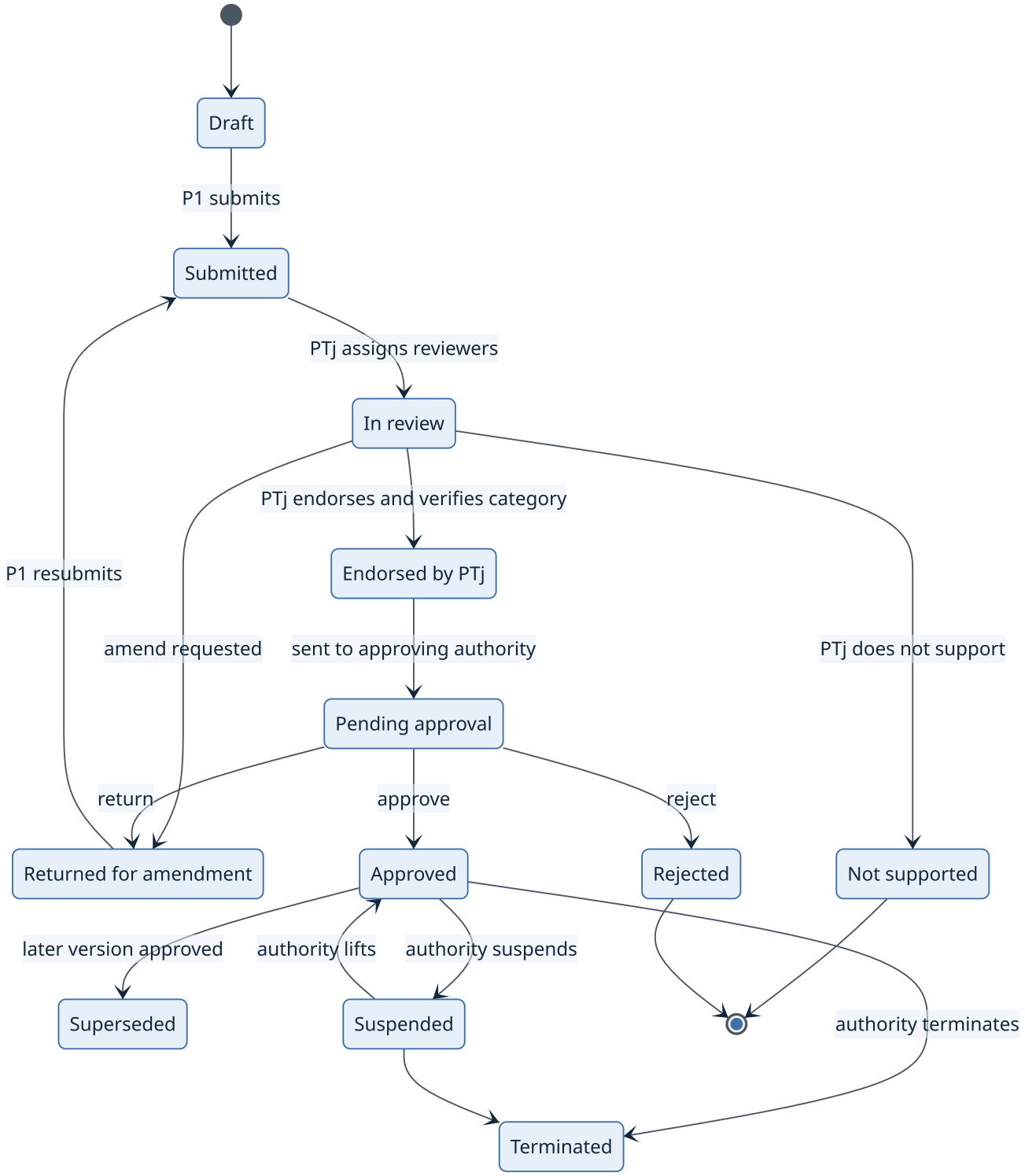


Figure 6: *States of a micro-credential version from draft to approval, and later supersession, suspension or termination.*

3.3.1 Academic Review

ID	Requirement	Source
FR-62	The PTj (P2) **SHALL** be able to assign one or more reviewers (X1) to a submitted proposal, with a due date and the scope of review.	G 9.4, 16.6

ID	Requirement	Source
FR-63	The review form **SHALL** cover: suitability of learning outcomes; alignment of learning and assessment; adequacy of SLT; level and scope of content; delivery mode; module structure; credential information; and compliance with academic and QA requirements. For a UMC it **SHALL** also cover evidence of mapping to the parent course. Each item **SHALL** be rated Satisfactory or Needs amendment, with comments.	G 9.4
FR-64	Each reviewer **SHALL** give a recommendation: Support, Amend or Do not support (Sokong / Pinda / Tidak Sokong).	App A9
FR-65	Each review comment **SHALL** be tracked with a status (open, answered, resolved) and the proposer's response. The proposal **SHALL NOT** go forward for approval while any comment marked as blocking is unresolved.	G 9.4
FR-66	The PTj **SHALL** explicitly verify the category when endorsing a proposal. The verified category **SHALL** be recorded separately from the proposed category.	App H3
FR-67	The system **SHALL** produce a review summary for the approving authority showing all recommendations, unresolved warnings and the proposer's responses.	G 9.5

3.3.2 Approval Workflow

ID	Requirement	Source
FR-68	The approval workflow **SHALL** record three stages, as in Appendix A9: academic or technical review, PTj or faculty endorsement, and approval by the UTeM authority configured for the decision type.	App A9, D-11
FR-69	Each decision **SHALL** record the decision-maker, their role, the decision (support / amend / not support, or approve / return / reject), the date, comments and an external reference such as a Senate meeting number.	G 15.7
FR-70	An approval **SHALL** refer to one exact version of the micro-credential. Approval of one version **SHALL NOT** carry over to any other version.	G 9.5
FR-71	Once a version is submitted its content **SHALL** be locked. Changes **SHALL** be possible only after it is returned for amendment, or through a new version.	G 9.5
FR-72	Approval to offer **SHALL NOT** set, imply or display any QV status, ABC status, credit transfer status or stackable status.	G 15.4, D-15, BR-16
FR-73	Approvers **SHALL** see a queue of items awaiting their decision with the age of each item. Items older than `alerts.decision_overdue_days` **SHALL** be reminded to the approver.	
FR-74	The approving authority **SHOULD** be able to record decisions for several items from one meeting in a single action with one meeting reference.	
FR-75	The system **SHALL** produce a decision pack per item (proposal summary, review summary, readiness, category verification) printable for meetings.	G 9.5

3.3.3 Material Change and Version Control

ID	Requirement	Source
FR-76	Every micro-credential **SHALL** have a version history. Each version **SHALL** record its number, effective date, change summary, approval decision and the implications for participants currently enrolled.	G 17.8, 18.8
FR-77	Every change request **SHALL** be classified as minor or material. The system **SHALL** classify a change as material automatically when it touches learning outcomes, SLT or credit, MQF level, core content, major assessment or pass standard, module structure, parent course link, QV status, a substantive delivery mode, the provider or partner, or the identity and integrity mechanism.	G 9.5, 11.2.3, 17.7, App A10

ID	Requirement	Source
FR-78	A material change **SHALL** create a new version that goes through review and approval again. It **SHALL NOT** take effect before it is approved.	G 17.7, BR-18
FR-79	A minor change **SHALL** be recorded with a description and impact note, acknowledged by the PTj, and logged in the CQI record of the offering.	G 17.7, App H7
FR-80	For a Quality-Verified SAMC, a material change **SHALL** be flagged as affecting QV and **SHALL** create a QV re-verification task (Section 3.10).	G 11.2.3, 17.7
FR-81	A version referenced by any participant record **SHALL** never be deleted.	G 18.8, BR-39
FR-82	Each enrolment **SHALL** be bound to the version under which the participant enrolled. When a new version takes effect, the PTj **SHALL** record the transition rule for current participants (finish on old version, move to new version, or equivalence).	G 10.3.2, 17.10
FR-83	When an approved change affects current participants, the system **SHALL** notify them of the change, the effective date and the transition rule.	G 10.3.2
FR-84	The system **SHOULD** show a side-by-side comparison of two versions with the changed fields highlighted.	

3.4 Offering, Catalogue and Readiness (M5)

An approved version is offered as one or more cohorts. Before registration opens, the PTj confirms that materials, platform, assessment, facilitators, registration, records and certification are ready, using the checklist of Guide Appendix B [G 9.6]. Participants must receive honest, complete information before they enrol [G 10.3.1].

3.4.1 Offering and Readiness

ID	Requirement	Source
FR-85	P1 or P2 **SHALL** be able to create an offering (cohort) for an approved version with: cohort name, start and end dates, delivery mode, capacity, registration window, fee, facilitators, eligible participant groups and the learning platform course reference.	App B1
FR-86	The system **SHALL NOT** allow an offering to be created for a version that is not in state Approved.	G 9.6, BR-17
FR-87	Each offering **SHALL** have a readiness checklist with the items of Appendix B2–B7 (transparency of information, materials and platform, assessment and integrity, participant and achievement records, completion and credential process, change notification), each with status Yes / Not yet / Not applicable and evidence notes.	App B
FR-88	Readiness **SHALL** be confirmed by the lecturer, a PTj representative and, where relevant, support or system staff, each choosing Ready, Ready with conditions or Not ready, as in Appendix B8.	App B8
FR-89	Registration **SHALL NOT** open until the PTj confirmation is Ready, or Ready with conditions where the conditions are recorded.	G 9.6
FR-90	The checklist **SHALL** be repeated for every new cohort, pre-filled from the previous cohort. If the check reveals a material change, the offering **SHALL** be held until the change is approved.	App B, B8 note
FR-91	For a SAMC that advertises Quality-Verified status, the system **SHALL** check that QV is valid until the end date of the cohort. If it is not, the offering **SHALL NOT** display a Quality-Verified claim, credit value or MQF level.	G 11.2.2, BR-23

3.4.2 Catalogue and Participant Transparency

ID	Requirement	Source
FR-92	The system **SHALL** provide a public catalogue, without login, of offerings that are open or upcoming.	G 9.6
FR-93	Each catalogue entry **SHALL** show at least: name, category, learning outcomes, content or modules, SLT, delivery mode, prerequisites, access period, assessment methods and standard, completion conditions, fees, learning support, the credential to be received, and, where applicable and valid, QV status, credit value and MQF level.	G 9.6, 10.3.1, App B2
FR-94	Users **SHALL** be able to search and filter the catalogue by category, PTj, field, delivery mode, language, start date and fee.	
FR-95	Recognition statements in the catalogue **SHALL** be generated from the approved wording for the category and status (Section 3.9.2). Free-text descriptions **SHALL** be checked for forbidden claims, such as a promise that credit will transfer or that a number of micro-credentials leads to a degree.	App E5, G 14.9, BR-22
FR-96	For an offering with credit potential, the catalogue entry **SHALL** show a standard explanation that MC completion, credential award, ACR preparation, ABC deposit and credit transfer approval are separate steps and that none guarantees admission, transfer or a qualification.	G 10.3.1, BR-26
FR-97	For a modular UMC, the catalogue entry **SHALL** show the module structure, how modules relate to the whole UMC, and whether staged completion is allowed.	G 9.6
FR-98	The catalogue **SHALL** be available in Bahasa Melayu and English.	D-02

3.5 Registration, Enrolment and Cohorts (M6)

Participants register for open offerings after reading the offering information. Fees are handled through the payment adapter; in Release 1 a dummy payment service records simulated payments (A-03). For a modular UMC, participants may complete modules in different periods, and each module status is kept separately [G 10.3, 12.2].

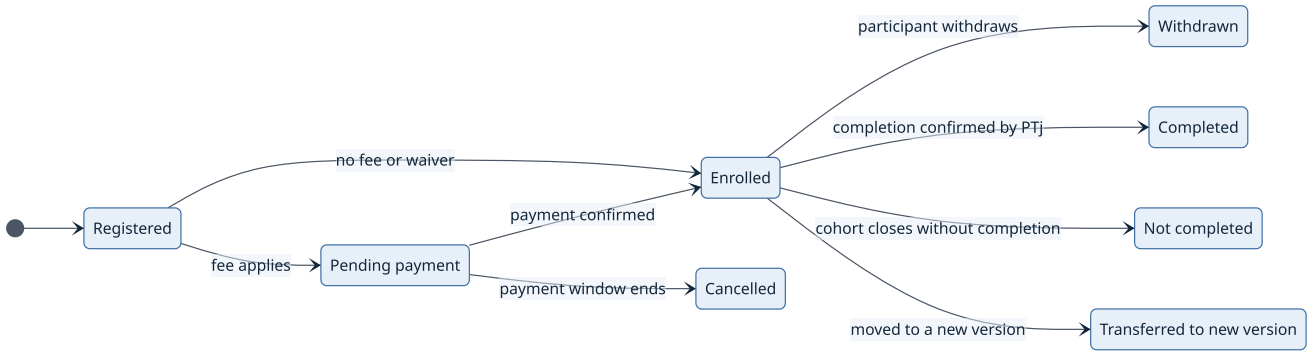


Figure 7: *States of an enrolment.*

ID	Requirement	Source
FR-99	A signed-in user **SHALL** be able to register for an offering whose registration window is open and whose capacity is not full.	G 10.3
FR-100	The system **SHALL** check the eligibility rules of the offering (participant group, prerequisites) and **SHALL** explain why a user is not eligible.	G 10.3.1
FR-101	Before confirming registration the participant **SHALL** acknowledge the offering information, including the recognition statements and the explanation of separate stages. The acknowledged text version **SHALL** be stored with the enrolment.	G 10.3.1

ID	Requirement	Source
FR-102	At first registration the system **SHALL** show the personal data notice and record the participant's acknowledgement.	PDPA
FR-103	Where a fee applies, the system **SHALL** request payment through the payment adapter. In Release 1 the adapter **SHALL** be a dummy service that records a simulated success or failure with a reference number.	D-14, A-03
FR-104	P2 **SHOULD** be able to grant a fee waiver or sponsorship to a participant with a recorded reason.	
FR-105	Each enrolment **SHALL** have one of the states of Figure 7. Every change of state **SHALL** be time-stamped and attributed.	G 10.3
FR-106	For a modular UMC that allows staged completion, the system **SHALL** record enrolment and completion status per module, with dates, even when modules are taken in different periods.	G 10.3, 12.2
FR-107	A participant **SHALL** be able to withdraw before the deadline set for the offering. Refunds are handled outside MCMS.	
FR-108	The system **SHOULD** keep a waiting list when an offering is full and offer released places in order.	
FR-109	P5 **SHOULD** be able to enrol a group of participants from a CSV file, for example a sponsored company group, with the same checks as individual registration.	
FR-110	P1 and P2 **SHALL** be able to view, filter and export the participant list of a cohort.	App H5
FR-111	The participant **SHALL** receive confirmation of registration, payment and enrolment, with access instructions for the learning platform.	G 10.2

3.6 Learning Platform Connection (M7)

Learning takes place on UCreds, the pilot learning platform. MCMS exchanges enrolments, progress, module completion, results and badge data with UCreds through the learning platform adapter. Because the connection method is not yet known, Release 1 uses CSV files (D-13, A-04, OI-04). Imported data is never final: it lands as provisional data awaiting verification (Section 3.7).

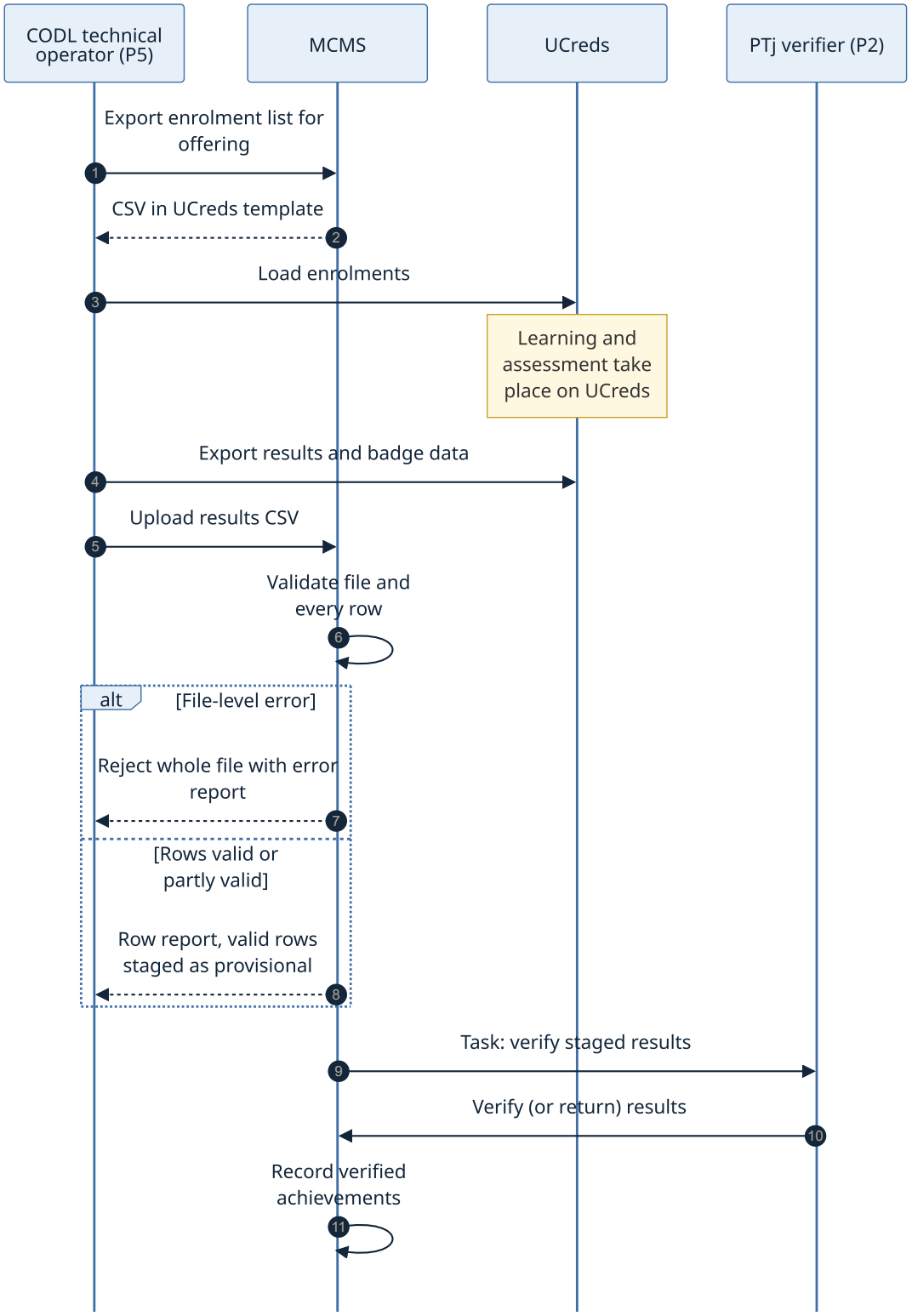


Figure 8: *Release 1 exchange with UCreds through CSV files. Imported results become achievements only after PTj verification.*

ID	Requirement	Source
FR-112	All data exchange with UCreds **SHALL** pass through the learning platform adapter interface, so that another platform or a live connector can replace the CSV implementation without changing other modules.	D-06, D-13
FR-113	The system **SHALL** export the enrolment list of an offering in a published CSV template for loading into UCreds.	A-04

ID	Requirement	Source
FR-114	The system **SHALL** import progress, module completion, assessment results and badge data from UCreds in published CSV templates.	A-04
FR-115	Every import **SHALL** be validated: file format and template version; each row matched to an enrolment, module and assessment component; score ranges; duplicates. A file-level error **SHALL** reject the whole file; row-level errors **SHALL** be listed in a downloadable report.	
FR-116	Imported results **SHALL** be stored as provisional and **SHALL NOT** create achievements, badges or credentials until verified under Section 3.7.	G 10.4, BR-20
FR-117	When UCreds issues module badges (A-05), the system **SHALL** import the badge identifier, link and date, match it to the verified module achievement, and flag any badge without a matching verified achievement.	A-05, G 8.7.1
FR-118	Every export and import **SHALL** be logged with file name, file hash, row counts, result, user and time. Uploaded files **SHALL** be kept for the retention period.	G 18.2
FR-119	The system **SHOULD** provide a live connector to UCreds (web service or Moodle interface) once the connection method is agreed.	OI-04
FR-120	The system **MAY** synchronise progress with the learning platform in near real time.	

3.7 Assessment and Achievement Verification (M8)

Recognition follows achievement, not attendance [G 6.1, 8.6]. Results are entered or imported, then verified by the owning PTj before any badge, certificate or Digital Credential is issued. Verification links the participant's identity, the assessment evidence and the result [G 10.4].

ID	Requirement	Source
FR-121	The assessment plan of each offering **SHALL** be taken from the approved version: components per module or per micro-credential, weighting, pass standard and linked learning outcomes. Facilitators **SHALL NOT** change it outside the change process of Section 3.3.3.	G 8.6, 10.4
FR-122	Facilitators (P1) **SHALL** be able to enter results per participant and component, or import them (Section 3.6). The system **SHALL** compute totals and pass or fail against the pass standard.	G 10.4
FR-123	The system **SHALL NOT** allow a pass, module completion or MC completion to be based only on attendance or participation records.	G 8.6, BR-19
FR-124	For each assessment the system **SHALL** record the identity verification and integrity mechanism applied. Facilitators **SHALL** be able to flag an integrity case; results under an open case **SHALL** be held and **SHALL NOT** be verified.	G 10.4, App B4, H7
FR-125	Results **SHALL** be verified by an authorised PTj user (P2) before they create achievements. Verification **SHALL** record the verifier, the date and the evidence that links the participant's identity, the assessment and the result.	G 10.4, BR-20
FR-126	The verifier **SHALL** be able to verify the results of a whole cohort or module at once, after reviewing a summary of pass rates and outstanding flags.	
FR-127	After verification, any change to a result **SHALL** require a reason and approval by P2. The previous value **SHALL** be kept and the change written to the audit trail. If a credential was already issued, the system **SHALL** open a credential correction task (Section 3.9.4).	G 18.10, D-20
FR-128	A module achievement **SHALL** be recorded when all learning outcomes and assessments of the module are met and verified.	G 12.2

ID	Requirement	Source
FR-129	A participant **SHOULD** be able to raise a query or appeal on a result. The case **SHALL** be tracked to a decision and linked to any result change.	G 15.8
FR-130	Participants **SHALL** be able to see their verified results and the feedback released by the facilitator.	App A6

3.8 Achievement Record (M9)

The Achievement Record (Rekod Pencapaian Pembelajaran Micro-Credential UTeM) is UTeM's internal record of modules completed, results, badges and MC completion. It is not an ACR and it is not evidence that credit was deposited to ABC or accepted by a programme [G 8.7.5, 18.3]. Figure 9 shows how a modular UMC builds up to completion.

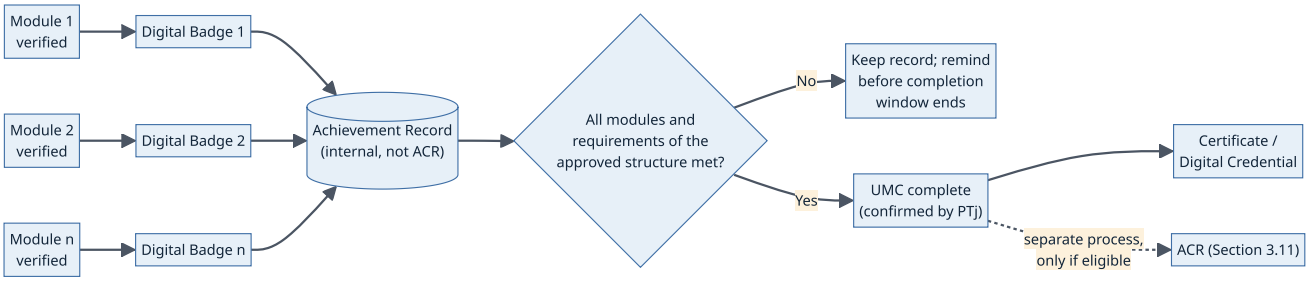


Figure 9: *Staged completion of a modular UMC [G 12.2, App E9]. A badge proves a module, not the UMC and not credit.*

ID	Requirement	Source
FR-131	The system **SHALL** maintain one Achievement Record per participant and micro-credential, holding modules completed, verified results, Digital Badges, MC completion status and credentials awarded.	G 8.7.5, 18.3
FR-132	The Achievement Record **SHALL** be stored separately from ACR records and **SHALL** be labelled on every screen and export as an internal record that is not an Academic Credit Record.	G 18.3, BR-25
FR-133	A UMC **SHALL** be marked complete only when every module and every requirement of its approved structure is met. The system **SHALL** calculate this automatically and **SHALL** require confirmation by P2.	G 8.7.3, 12.3.1
FR-134	The system **SHALL NOT** combine badges or modules from different UMCs or different parent courses to complete a UMC. A module from another structure **SHALL** count only through a recorded equivalence decision by the academic authority.	G 8.7.2, 12.2, BR-12
FR-135	For a modular UMC the system **SHALL** track the completion window (default `umc.completion_window_months`, which has no value until OI-13 is answered; an offering **MAY** set its own period) and **SHALL** remind the participant before it ends. While no period is set, no reminder is sent.	G 7.4, 12.2, OI-13
FR-136	A SAMC or MCE **SHALL** be marked complete only when all learning outcomes, SLT activities, assessments and completion conditions are met and verified, and the completion is confirmed by P2.	G 12.3.2, 8.7.4
FR-137	The participant **SHALL** be able to see their own Achievement Record, including progress across modules taken in different periods.	G 10.3
FR-138	P1 and P2 **SHALL** be able to see the Achievement Records of participants in their offerings.	App H4

3.9 Credential Issuance and Verification (M10)

MCMS issues three kinds of proof of achievement. They have different functions and must never be used in place of each other [App E1]. Credentials are issued in an open, signed format so that anyone can verify them without contacting UTeM (D-19).

Table 8: *Types of proof of achievement (Guide Appendix E1)*

Type	Main function	Level of achievement	Relation to credit
Digital Badge	Proof of achievement of a module, unit or milestone	Usually a module or learning unit	Not academic credit by itself
Certificate (Sijil)	Proof of completing the offering under its conditions	Whole micro-credential	Does not guarantee credit or transfer
Digital Credential	Verifiable digital record describing the achievement, the issuer and related information	Whole micro-credential or a set achievement	May state credit or MQF level only when verified; does not guarantee transfer

3.9.1 Issuance

ID	Requirement	Source
FR-139	Before any credential is issued the system **SHALL** run the checks of Appendix E8 automatically: identity matches the official record; all completion conditions are met; results are verified; the credential type fits the level of achievement; the title and category match the approved version; any credit, MQF or QV information is verified and still valid; the verification mechanism works. Issuance **SHALL** be blocked if any check fails.	App E8, BR-21
FR-140	The system **SHALL** issue a Digital Badge for each verified module achievement when the approved design includes badges and UCreds does not issue them (A-05). A badge **SHALL** contain: badge name, parent micro-credential, issuer (UTeM or the authorised PTj), recipient, issue date, achievement criteria, learning outcomes or competencies in its metadata, and a verification link.	App E2, A-05
FR-141	A Digital Badge **SHALL NOT** be described or displayed as academic credit.	App E2, BR-11
FR-142	The system **SHALL** issue a Certificate only after the whole micro-credential is complete. A Certificate **SHALL** contain: the official name of UTeM or the provider; the participant's name as in the official record; the approved title; the category where needed for clarity; a statement of completion; the award date; a unique certificate number; the authorised signatory; credit, QV and MQF information only when applicable, verified and valid; and a QR code for verification.	App E3, G 10.5
FR-143	The system **SHALL** issue a Digital Credential with the metadata of Appendix E4: unique credential ID; credential name; recipient; issuer; issue date; learning outcomes or competencies; learning time (SLT); assessment basis; credit value and MQF level only when verified; QV status only after QV is obtained; expiry date of a related status where required; and the verification mechanism.	App E4
FR-144	Digital Credentials **SHALL** follow the Open Badges 3.0 and W3C Verifiable Credentials data models as guidance, **SHALL** be digitally signed with a UTeM key (VC-JOSE, Ed25519), and **SHALL** be downloadable as signed JSON (`.json`), signed token (`.jwt`), and PDF rendering files. Blockchain **SHALL NOT** be required.	D-19, D-36, C8
FR-145	Credit value, MQF level and QV wording on any credential **SHALL** be filled only from verified records: the approved UMC credit, or the QV decision for a SAMC. These fields **SHALL NOT** be editable by hand.	D-17, C5, BR-22

ID	Requirement	Source
FR-146	The system **SHALL** issue credentials automatically, for one participant or a whole cohort, as soon as the deputy dean (P2) has verified the results or confirmed completion and the pre-issuance checks pass. No separate manual issuing step **SHALL** be required; participants who fail a check **SHALL** be listed for the deputy dean with the reason.	App H3, D-29
FR-147	Certificates and Digital Credentials **SHALL** be available in Bahasa Melayu and English.	D-02, C7
FR-148	Every credential **SHALL** have a unique, non-guessable identifier formatted as a hyphenated Crockford Base32 code (e.g. `OOP01-3Q7SX`), and a canonical verification URL encoded as a high-resolution QR code (exportable at 3x scale PNG and vector SVG). Verification URLs **SHALL** dynamically resolve to the active host and port in local development and staging environments.	App E6, D-40, D-41
FR-149	The participant **SHALL** be able to view and download their credentials and copy a share link.	
FR-150	The participant **SHOULD** be able to add a credential to professional networking profiles directly.	
FR-151	The system **MAY** export credentials to external digital credential wallets.	

3.9.2 Recognition Wording Control

Credentials and offering information must not suggest credit, MQF level or QV status that has not been formally obtained [G 10.5, App E5]. The system therefore builds recognition statements from controlled templates.

Table 9: *Allowed and forbidden recognition statements (Guide Appendix E5)*

Situation	Allowed statement	Statement to avoid
UMC	Link to the parent course or programme and the approved credit information under academic rules	"Credit will certainly transfer"; any promise of admission to a programme
SAMC before QV	SLT, and that QV is in progress or not applied for, if needed	Credit value or MQF level presented as verified
Quality-Verified SAMC	Credit value, MQF level and QV status as verified and still valid	That QV guarantees credit transfer
MCE	Completion of learning or competency, and suitable duration or SLT	MCE as academic credit or as a direct credit transfer route

ID	Requirement	Source
FR-152	The system **SHALL** generate every recognition statement on credentials, catalogue entries and participant notices from the template for the confirmed category and current status in Table 9.	App E5, BR-22
FR-153	The system **SHALL** block issuing or publishing any text that states a credit value, MQF level or QV status not supported by a verified and valid record.	G 10.5, D-17
FR-154	When QV expires or is suspended or withdrawn, credentials issued afterwards **SHALL** carry no Quality-Verified claim. Verification of an earlier credential **SHALL** show the QV status that was valid on its issue date.	G 11.2.2, BR-23

3.9.3 Public Verification

ID	Requirement	Source
FR-155	The system **SHALL** provide a public verification page, without login, reached by credential ID, verification URL or QR code.	App E6, D-19
FR-156	The verification page **SHALL** present an authoritative certificate-style result slip featuring a single prominent status seal (e.g. "Status: Valid"), a copyable verification code, a structured two-column key-value summary, direct verifier download actions for `.json` and `.jwt` cryptographic files, print slip styling, and an explicit statutory Personal Data Protection Act 2010 (PDPA) notice. It **SHALL NOT** expose identity card or passport numbers, matric numbers, email addresses or results.	App E6, PDPA, D-43
FR-157	The verification page **SHOULD** let a verifier upload a Digital Credential file and check its signature and integrity against the issuer key.	D-19
FR-158	Machine-to-machine public key discovery **SHALL** be published at `/.well-known/jwks.json` adhering to RFC 7517. It **SHALL NOT** be rendered as an interactive button within public navigation headers, and **SHALL** be linked from the institutional trust framework.	D-40
FR-159	Each verification request **SHALL** be logged with time, credential ID and outcome, without storing the verifier's personal data beyond a network address kept for security purposes.	App E6
FR-160	A request for an unknown or malformed credential ID **SHALL** return a neutral "not found" message that does not reveal whether a similar ID exists.	

3.9.4 Correction, Replacement and Revocation

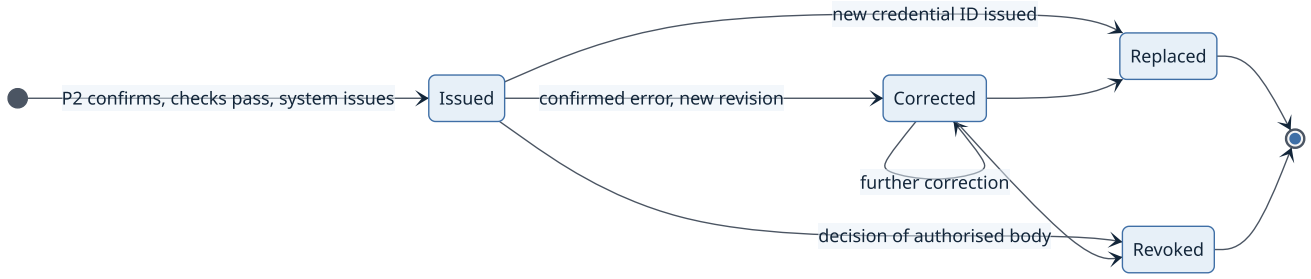


Figure 10: *Life of an issued credential [App E7]. No state change deletes the earlier record.*

ID	Requirement	Source
FR-161	A credential **SHALL** be corrected only for a confirmed error, by the deputy dean (P2) of the owning PTj. The record **SHALL** keep the original content, the reason, the correction, the persons involved and the date.	App E7, G 10.5.1
FR-162	Replacement **SHALL** issue a new credential with a new ID linked to the old one. Verification of the old ID **SHALL** show "Replaced" and the new ID.	App E7
FR-163	Revocation **SHALL** be possible only through a recorded decision of an authorised body with a stated reason. Verification **SHALL** show "Revoked" and the date.	App E7
FR-164	No credential record or earlier revision **SHALL** ever be deleted.	G 10.5.1, 18.4, BR-38
FR-165	The participant **SHALL** be notified of any correction, replacement or revocation of their credential, with the reason.	G 18.10

3.10 Quality Verification Management for SAMC (M11)

Quality Verification (QV) is the quality check for a SAMC carried out by the QVC under MQA rules. It is separate from UTeM's approval to offer: approval lets a micro-credential be offered; QV confirms its quality and gives it a credit value and MQF level [G 11.2]. QV requires at least one completed cohort and is valid for at most three continuous years [G 11.2.1–11.2.2]. In Release 1, MCMS prepares and tracks; UTeM staff submit to QVC/MQA outside the system (D-14).

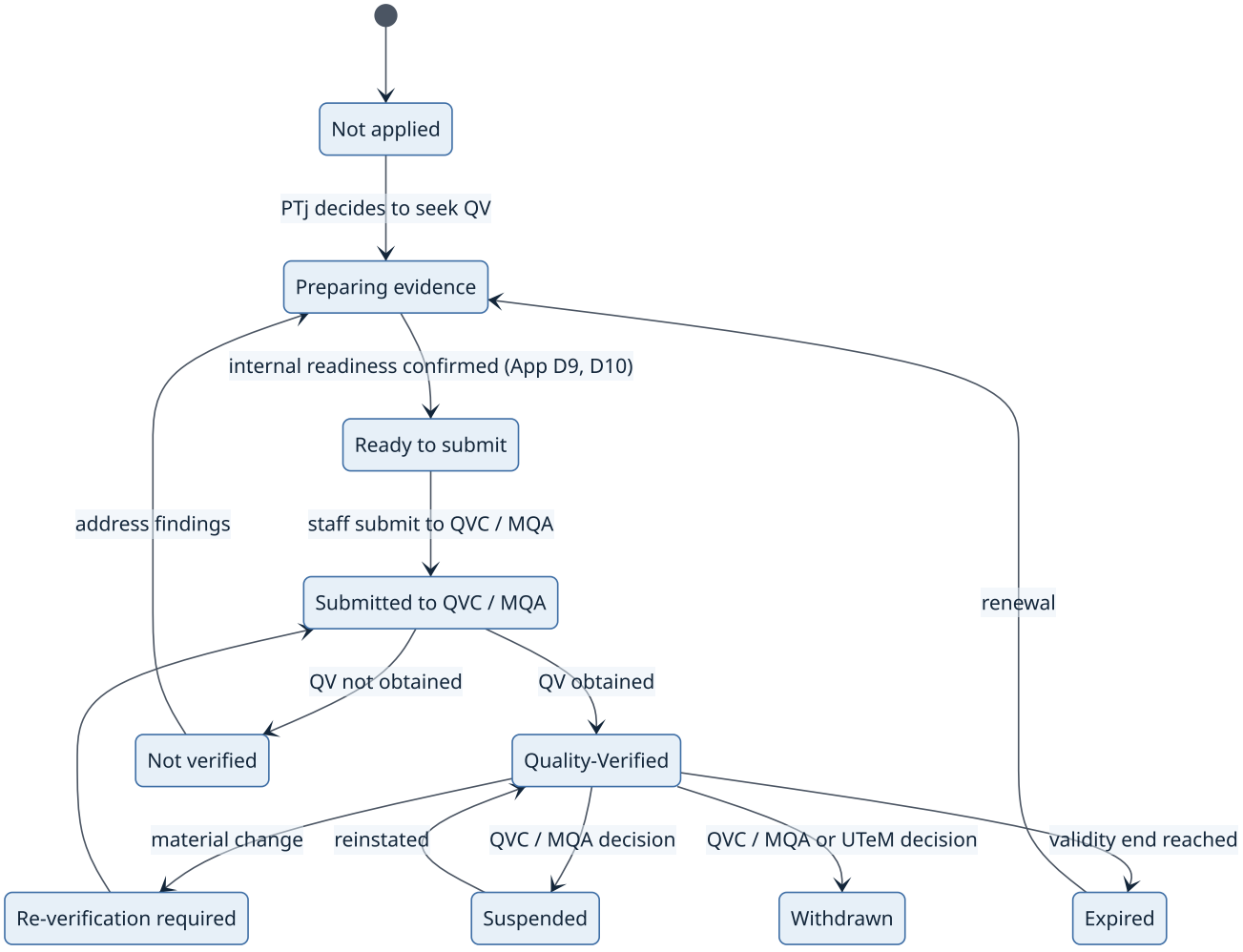


Figure 11: *QV status of a SAMC. Only the state Quality-Verified allows credit value, MQF level and QV claims.*

ID	Requirement	Source
FR-166	The QV functions **SHALL** be available only for micro-credentials whose confirmed category is SAMC.	G 7.6, 11.2
FR-167	The system **SHALL** allow the status "Ready to submit" only when the basic conditions of Appendix D2 are met: the SAMC is a stand-alone offering; total SLT is at least `slt.min_hours_samc`; the provider is registered and authorised; at least `qv.min_completed_cohorts` cohort has completed with complete evidence; and any proposed credit value and MQF level are justified and marked as proposed.	G 11.2.1, App D2, BR-03
FR-168	The system **SHALL** keep an evidence checklist with the items of Appendix D3–D7 (academic design, assessment and achievement, delivery and participant support, quality assurance and CQI, Digital Credential), each with status Available / Not yet / Not applicable, file reference and notes.	App D

ID	Requirement	Source
FR-169	The system **SHALL** link evidence that already exists in MCMS automatically: learning outcomes and mapping, SLT calculation, assessment plan, cohort results and achievement analysis, feedback, CQI records, version history and credential design.	App D3–D7
FR-170	The system **SHALL** keep the material change review of Appendix D8 for each QV cycle, listing for each component whether it changed and whether it needs internal review or referral to QVC/MQA.	App D8, G 11.2.3
FR-171	The internal readiness decision of Appendix D9 (ready; ready with minor corrections; not ready), the list of missing critical evidence and the target submission date **SHALL** be recorded by the lecturer and confirmed by the deputy dean (Appendix D10; D-28).	App D9, D10
FR-172	The system **SHALL** produce a submission package: an evidence index, a summary document and a compressed file of the evidence, for staff to submit to QVC/MQA.	D-14, A-02
FR-173	Staff **SHALL** record the submission date, the QVC/MQA reference and the person who submitted.	G 11.2
FR-174	Staff **SHALL** record the QV decision: verified or not verified, QV reference number, verified credit value, verified MQF level, start date, expiry date and any conditions, with the decision document attached. The system **SHALL** reject an expiry date more than `qv.max_validity_years` after the start date.	G 11.2.2, BR-02
FR-175	The verified credit value and MQF level in the QV record **SHALL** be the only source of these values for catalogue entries, credentials and ACR of the SAMC.	D-17, BR-14
FR-176	The system **SHALL** alert the lecturer (P1) and deputy dean (P2) at each interval in `alerts.qv_expiry_days` before QV expires.	G 11.2.2, D-16
FR-177	When QV expires, is suspended or is withdrawn, the system **SHALL** immediately stop Quality-Verified claims in the catalogue and on new credentials, **SHALL** stop new ACR eligibility for completions after that date, and **SHALL** create a task for the PTj to manage affected participants and records.	G 11.2.2, App H7, BR-23
FR-178	A QV status **SHALL NOT** set or imply any credit transfer, admission or stackable status.	G 11.2, 15.4, BR-16
FR-179	The system **MAY** submit QV applications to QVC/MQA electronically when such an interface exists.	A-02

3.11 ACR and ABC Deposit (M12)

For eligible learning, an Academic Credit Record (ACR) can be prepared and deposited to the Academic Bank of Credit (ABC). Only two sources are eligible: a completed UMC from an accredited programme (formal learning) and a completed Quality-Verified SAMC (non-formal learning) [G 12.1.1]. The deposit is storage, not approval: credit in ABC still has to be evaluated by a receiving programme before it can be used [G 12.5]. The participant's consent is required [G 12.4.2]. Figure 12 shows the Release 1 process, in which the system generates the ACR, the deputy dean confirms it and, after the participant consents, records the deposit; the national ABC platform is still a concept, so no live connection exists yet (D-14, D-29).

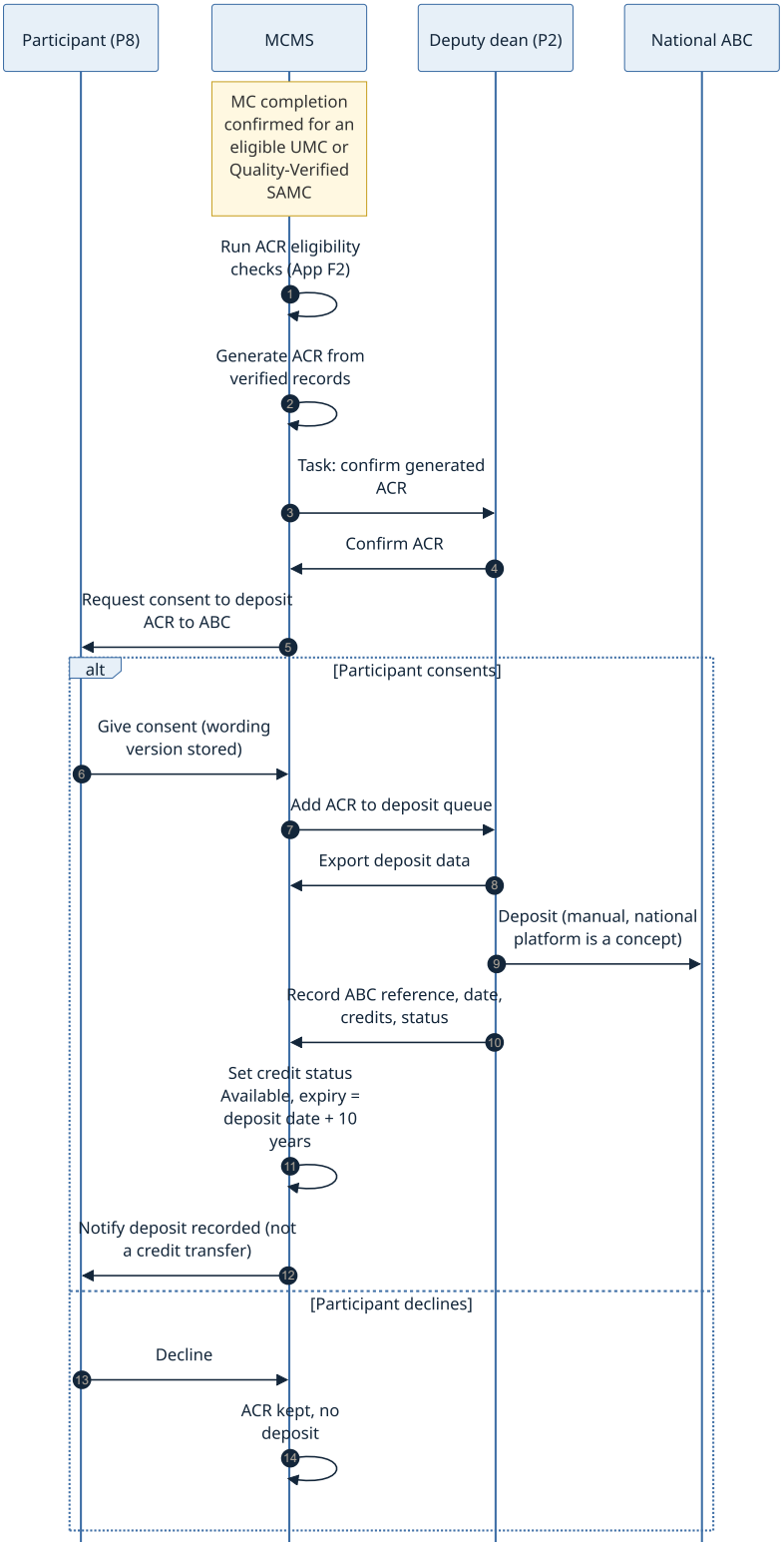


Figure 12: *Preparing an ACR and recording its deposit to ABC in Release 1.*

3.11.1 ACR Eligibility and Preparation

ID	Requirement	Source
FR-180	The system **SHALL** allow an ACR only for (a) a completed UMC from an accredited programme that meets the deposit requirements, or (b) a completed Quality-Verified SAMC whose QV was valid on the completion date. It **SHALL NOT** allow an ACR for an MCE, a SAMC without valid QV, a single module, a badge or a certificate.	G 12.1.1, 12.5, BR-24

ID	Requirement	Source
FR-181	Before an ACR is prepared the system **SHALL** run and record the checks of Appendix F2: MC completed; identity and results verified; UMC from an accredited parent programme where relevant; valid Quality-Verified status for a SAMC where relevant; credit value and MQF level based on a valid approval or QV decision; source records and credential traceable.	App F2
FR-182	The system **SHALL** generate the ACR from verified records with the fields of Appendix F3 and clause 12.4.2: ACR reference, date created, participant identity, provider or issuer, course or credential name, learning outcomes, credit value, MQF level, completion date, assessment basis and verification metadata. These fields **SHALL NOT** be typed by hand.	G 12.4.2, App F3, D-17
FR-183	The deputy dean (P2) of the owning PTj **SHALL** confirm each generated ACR before it can be offered for deposit. The system is Responsible for generating it and P2 Accountable for confirming it.	App F3, H3, D-29
FR-184	ACR **SHALL** be a separate record type from the Achievement Record and from credentials, and **SHALL** be clearly labelled as such on every screen and export.	G 12.4, 18.5, D-15
FR-185	A participant **SHALL** be able to hold several ACRs. Each ACR **SHALL** keep its own identity and academic information.	G 12.6
FR-186	The system **SHALL NOT** combine credits or modules from different ACRs to form a UMC, course or programme component without a recorded equivalence decision.	G 12.6
FR-187	The system **SHOULD** let staff record credit records from other institutions that a participant presents for transfer, marked as external, with their provider, QV or accreditation status, credit, MQF level and evidence.	G 12.7

3.11.2 Consent and Deposit Tracking

ID	Requirement	Source
FR-188	The system **SHALL NOT** place an ACR in the deposit queue without the participant's explicit consent. The consent record **SHALL** hold the date, the wording version and the channel.	G 12.4.2, App F4, BR-27
FR-189	The participant **SHALL** be able to decline or postpone consent. The ACR **SHALL** be kept and can be deposited later if the participant consents.	G 16.8
FR-190	The deputy dean (P2) **SHALL** have a queue of consented, confirmed ACRs awaiting deposit.	App H3, D-28
FR-191	The system **SHALL** export the deposit data of selected ACRs in a file format agreed with ABC (TBD, OI-11).	D-14
FR-192	The deputy dean (P2) **SHALL** record for each deposit: deposit date, ABC reference, credits deposited, status (Successful, In process, Correction needed) and the person recording it.	App F4
FR-193	The system **SHALL** NEVER display or record an ABC deposit as an approved credit transfer. Deposit screens and notices **SHALL** state that deposit is storage only.	G 12.5, 18.6, BR-26
FR-194	Badges, certificates, Digital Credentials and module records **SHALL NOT** be selectable for deposit.	G 12.5
FR-195	The system **MAY** deposit ACR to ABC through a live interface when ABC provides one.	A-02

3.11.3 Credit Validity, Ledger and Retention

ID	Requirement	Source
FR-196	On recording a deposit the system **SHALL** compute the credit expiry date as the deposit date plus `credit.abc_validity_years` and **SHALL** show it to the participant.	G 12.8, BR-04
FR-197	The system **SHALL** alert the participant at each interval in `alerts.credit_expiry_days` before credit expires.	D-16
FR-198	After the expiry date the credit **SHALL** show as Expired for transfer purposes. The ACR itself **SHALL** remain. A later transfer request using expired credit **SHALL** be flagged for re-evaluation, including currency of knowledge.	G 12.8, BR-28
FR-199	The participant **SHALL** have a credit ledger in the layout of Appendix F5 listing, per ACR: credential, provider, credits, deposit date, expiry date and status (Available, Provisionally Redeemed, Redeemed, Expired), with a total and a note that accumulated credits are not automatically accepted by any programme.	App F5
FR-200	ACR **SHALL** be retained indefinitely unless the participant as record owner, or a recognised authority, directs otherwise. Such a direction **SHALL** be recorded and carried out only by an authorised user.	G 12.8, BR-28

3.12 Credit Transfer (M13)

Credit transfer is the evaluation and acceptance, by the programme that receives it (in MCMS, the deputy dean of the faculty that owns the degree, D-11), of valid credit to meet part of its requirements [G 13]. It is separate from accumulation: credit that is verified, recorded in an ACR or deposited in ABC must still be evaluated. Equivalence is judged on learning outcomes and related criteria, never on credit value or SLT alone [G 13.3].

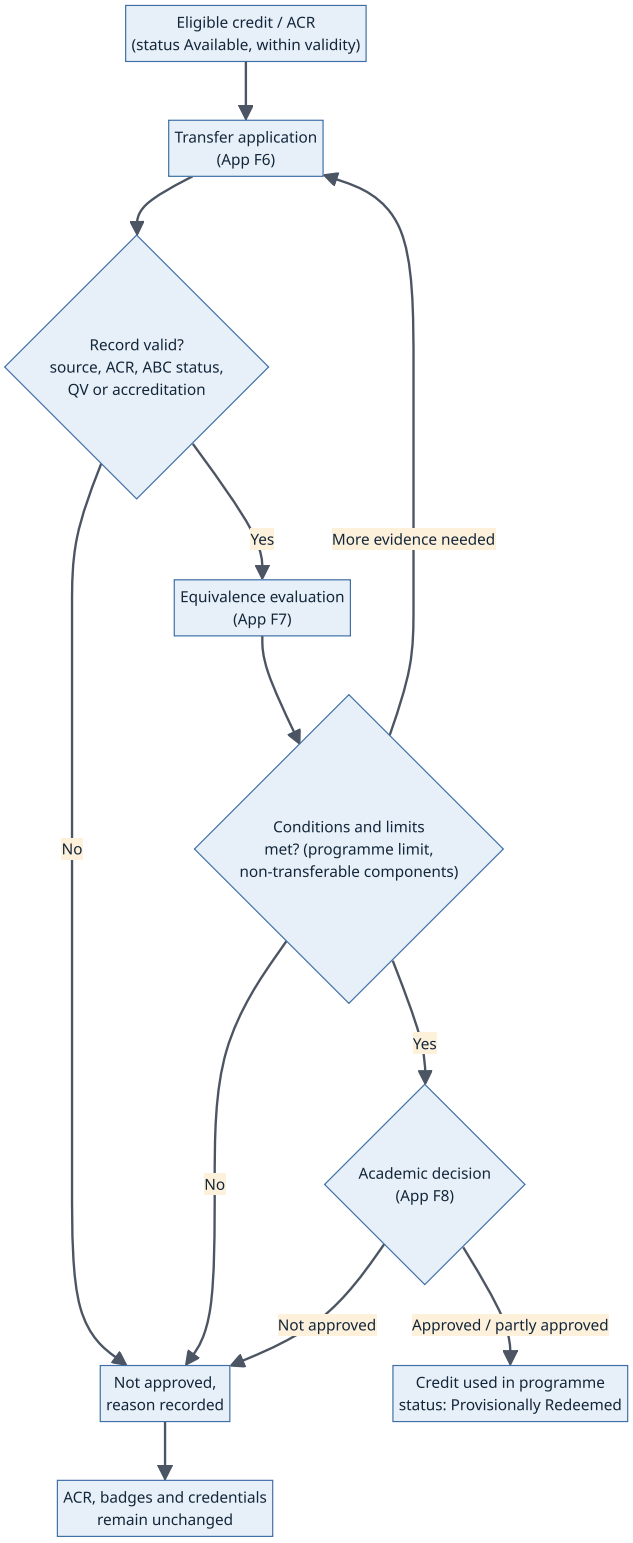


Figure 13: *Credit transfer process [G 13.4, Figure 14 of the Guide].*

ID	Requirement	Source
FR-201	A participant, or staff on the participant's behalf, **SHALL** be able to submit a transfer application with the fields of Appendix F6: target programme, receiving institution, course or component requested, credits requested, application date, the ACR or credits used, and supporting documents (ACR, credential, learning outcomes, content, SLT, assessment, QV evidence, other).	App F6

ID	Requirement	Source
FR-202	The system **SHALL** accept as sources only a UMC of UTeM, a UMC of another eligible institution, or a Quality-Verified SAMC with verifiable status, credit and MQF level. An application using an MCE **SHALL** be refused with an explanation.	G 13.2, BR-32
FR-203	Only credit with status Available **SHALL** be selectable. Expired credit **SHALL** be selectable only with a flag requiring re-evaluation.	G 12.8, 12.9
FR-204	The evaluator **SHALL** record a check of the validity of the record: authenticity of the ACR or credential, ABC status, and QV or accreditation status.	G 13.4
FR-205	The deputy dean (P2) of the faculty that owns the target degree **SHALL** complete an equivalence evaluation with the criteria of Appendix F7: learning outcomes; credit value and SLT; MQF level; content and depth; learning activities; assessment and standard; accreditation or QV status; currency of knowledge; programme or professional body requirements. Each criterion **SHALL** have a finding and a result.	G 13.3, App F7
FR-206	The system **SHALL NOT** allow the decision Approved for a course or component where the learning outcome criterion is recorded as Not equivalent. Matching credit value or SLT alone **SHALL NOT** be accepted as evidence of equivalence.	G 13.3, BR-30
FR-207	The system **SHALL** check the application against the transfer limit of the target programme and its list of non-transferable components (programme settings, TBD, OI-09), and **SHALL** warn the decision-maker of any breach.	G 13.5
FR-208	A partly completed UMC, or a set of badges, **SHALL NOT** be accepted for full credit of the parent course, unless a module carries an approved "separately recognisable credit unit" flag.	G 13.5, D-18, BR-13
FR-209	The decision **SHALL** be recorded with the fields of Appendix F8: Approved, Partly approved, Not approved or More evidence needed; credits approved; course or component credited; decision date; deciding authority; conditions or justification.	G 13.6, App F8
FR-210	Only a user holding the role configured as the transfer decision authority for the target programme **SHALL** record the decision.	App H3, D-11, A-01
FR-211	A decision of Not approved **SHALL NOT** change the ACR, Digital Badges, Certificates or Digital Credentials, and **SHALL NOT** change the credit status.	G 13.6, BR-33
FR-212	Each decision **SHALL** be traceable to the credit source, the ACR, the equivalence evaluation, the approval and the programme record.	G 13.7
FR-213	Each equivalence evaluation **SHALL** record the curriculum version of the target programme and the version of the micro-credential. When either has a material change, earlier equivalence results **SHALL** be flagged for review before reuse.	G 13.8, 14.8
FR-214	The system **SHALL** treat programme admission as a separate fact from transfer approval. Approval of a transfer **SHALL NOT** mark a participant as admitted.	G 14.4, BR-34
FR-215	The system **SHOULD** keep a library of approved equivalence mappings (micro-credential to course) that evaluators can consult. A precedent **SHALL NOT** replace the individual decision.	G 14.7

3.13 Single Redemption of Credit (M12)

Credit that has been used toward one qualification cannot be used toward another. Transferred credit is *provisionally redeemed* until the participant completes the programme; it then becomes *redeemed*. If the participant does not complete the programme, the credit can become available again while it is still valid [G 12.9].

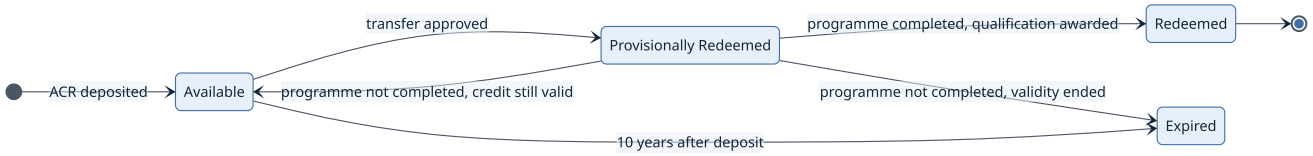


Figure 14: *Credit status under the Single Redemption Principle [G 12.9, App F9]. Redemption never deletes the ACR or the credentials.*

ID	Requirement	Source
FR-216	When a transfer is approved, the credits used **SHALL** change to Provisionally Redeemed and **SHALL** be linked to the programme and the participant's programme record.	G 12.9, App F9, BR-29
FR-217	When completion of the programme is recorded, the linked credits **SHALL** change to Redeemed.	G 12.9
FR-218	When the participant's withdrawal or non-completion of the programme is recorded, the linked credits **SHALL** return to Available if still within validity, or change to Expired otherwise, subject to ABC rules.	G 12.9
FR-219	Credit in status Provisionally Redeemed or Redeemed **SHALL NOT** be selectable in any other transfer application. The system **SHALL** explain why.	G 12.9, BR-29
FR-220	The system **SHALL** block submission of a second active application that uses the same ACR credits as an application still under evaluation.	G 12.9
FR-221	Redemption **SHALL NOT** delete or alter the ACR, Digital Badges, Certificates or Digital Credentials.	G 12.9
FR-222	In Release 1, programme completion and non-completion **SHALL** be recorded manually by an authorised user from the programme's records. Each status change **SHALL** record the evidence reference.	G 14.10, OI-10
FR-223	For each status change the system **SHALL** record whether the matching update was made in ABC (manual in Release 1), with date and reference.	G 12.9, 18.6

3.14 Stackable Degree Pathway (M13)

A stackable degree pathway lets eligible credit from micro-credentials count toward part of a degree. It is not a sum of credits. The participant must be admitted to the programme, have credit approved through transfer, meet the minimum of 30% of graduating credit earned at UTeM, meet residency for conventional programmes, cover the programme learning outcomes and Body of Knowledge, and meet all graduation requirements [G 14]. MCMS supports two levels: a **pathway mapping** for a target programme, approved by the academic authority, and an **individual stackable plan** per participant, using Guide Appendix G.

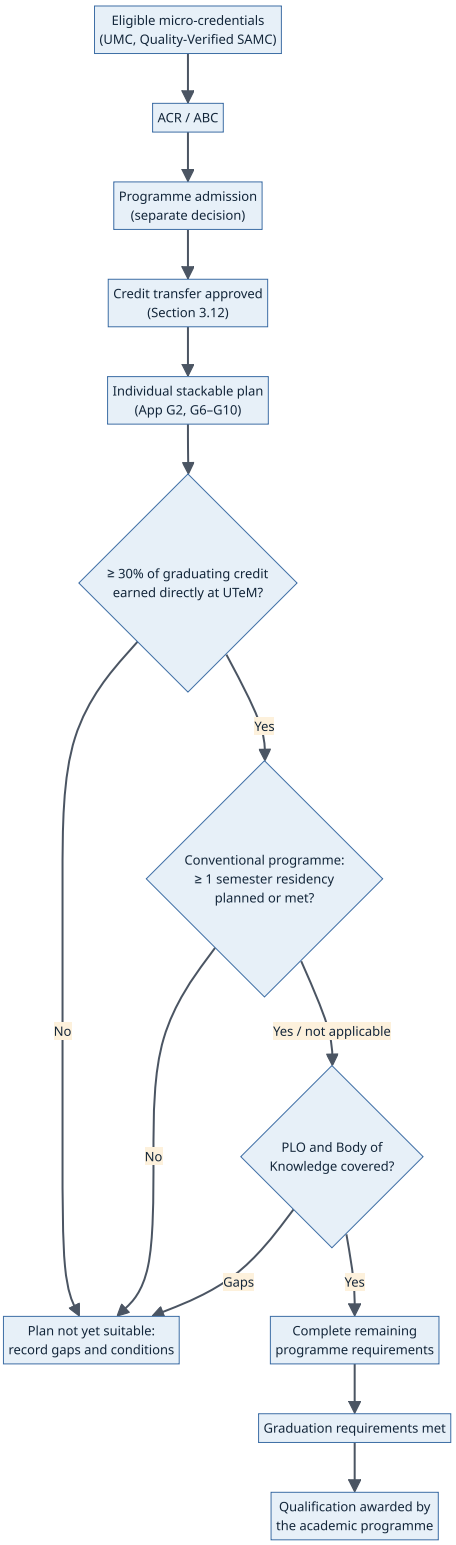


Figure 15: *Stackable degree pathway [G 14.3, Figure 16 of the Guide]. Admission, transfer and award are separate decisions.*

3.14.1 Pathway Mapping for a Programme

ID	Requirement	Source
FR-224	P1 or P2 **SHALL** be able to define a pathway mapping for a target programme via an accessible multi-step modal dialog ("+ Define mapping") with the information of Appendix G1: programme name and code, MQF level, total graduating credits, approved mode (conventional, ODL, other), faculty, accreditation status and intake.	App G1, D-44

ID	Requirement	Source
FR-225	The mapping **SHALL** list each micro-credential or credit source and the target course or component it maps to, with credits, learning outcome and content equivalence (High, Partial, None) and a preliminary decision (Suitable, Needs review, Not suitable).	App G3, G 14.7
FR-226	Only credit-bearing credentials (UMC, Quality-Verified SAMC, prior qualifications with credit) **SHALL** be mapped. An MCE **SHALL NOT** be mapped.	G 14.1.1, BR-35
FR-227	A pathway mapping **SHALL** be approved by the authority configured for stackable mapping before it is published to participants.	App H3, D-11
FR-228	An approved mapping **SHALL NOT** itself transfer any credit. Each participant's credit **SHALL** still go through Section 3.12.	G 14.7
FR-229	The mapping **SHOULD** include the PLO matrix of Appendix G4 (credential, PLO, contribution as main or supporting, evidence, gap) and the Body of Knowledge matrix of Appendix G5 (domain, requirement, contributing credential, coverage, gap). The system **SHOULD** warn when any PLO or domain has no coverage.	G 14.5.2, App G4, G5, BR-36
FR-230	When the target programme curriculum changes materially, the system **SHALL** flag the mapping for review.	G 14.8

3.14.2 Individual Stackable Plan

ID	Requirement	Source
FR-231	P1 or P2 **SHALL** be able to create an individual stackable plan for a participant with the content of Appendix G2 and G6–G10: the credentials and credits proposed, the UTeM credit check, the residency check, the remaining programme requirements, the credit and gap summary and the decision.	App G
FR-232	The system **SHALL** calculate the share of graduating credit to be earned directly at UTeM and **SHALL** mark the plan as not meeting the requirement when it is below `stackable.min_home_credit_percent`. The system **SHALL** state that meeting 30% does not mean 70% can be transferred automatically.	G 14.5.1, App G6, BR-05
FR-233	For a conventional programme the system **SHALL** require at least `stackable.residency_semesters` of residency to be planned or met, and **SHALL** record the proposed semester and delivery method. For a delivery method that may enter ODL scope, the plan **SHALL** be flagged for review under ODL policy.	G 14.5.3, 14.5.4, App G7, BR-06
FR-234	The plan **SHALL** list the remaining programme requirements (core, elective, project or capstone, industrial training, university requirements, professional body requirements) with status and planned completion.	G 14.5, App G8
FR-235	The plan decision **SHALL** be one of Suitable to proceed, Suitable with conditions, Not yet suitable or Not suitable, with conditions and follow-up actions, confirmed as in Appendix G11.	App G10, G11
FR-236	The system **SHALL NOT** allow the plan decision Suitable to proceed unless programme admission is recorded.	G 14.4, BR-34
FR-237	The award of a qualification **SHALL** be recorded only by an authorised user from the programme's records after admission and all requirements are met. MCMS **SHALL NOT** award qualifications.	G 14.10, BR-37
FR-238	Participants **SHALL** see accurate information about the pathway: target programme, mapping, admission requirements, transfer process, credit limits, credit validity, remaining requirements and graduation conditions. No screen **SHALL** state or imply that a number of micro-credentials or badges leads to a degree automatically.	G 14.9
FR-239	The system **MAY** read admission, enrolment and graduation status from the student information system through an adapter.	OI-10

3.15 Monitoring and Continuous Quality Improvement (M14)

Each offering is monitored with evidence, especially learning outcome attainment, assessment performance, completion and participant experience; the findings drive CQI [G 11.3–11.4, 17]. CQI is complete only when the effect of an action has been evaluated [G 17.12].

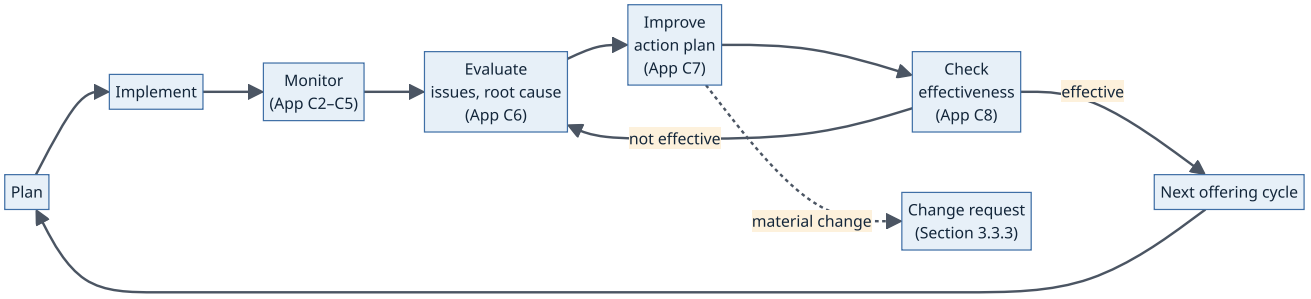


Figure 16: *CQI cycle [G 17.6, 17.12]. A CQI action that changes key academic components goes through re-approval.*

3.15.1 Monitoring

ID	Requirement	Source
FR-240	The system **SHALL** compute for each cohort the indicators of Appendix C2: active participation rate, completion rate, pass rate, average and median achievement, credential issuance rate, and significant access or technical issues, each compared with a target set for the offering.	App C2
FR-241	The system **SHALL** compute learning outcome attainment per outcome (share of participants who achieved it) from the mapping of assessment to outcomes.	G 17.5, App C3
FR-242	The system **SHALL** collect participant feedback after completion with a configurable questionnaire covering content, activities, assessment, platform, support and relevance. It **SHOULD** also collect facilitator and industry feedback.	G 17.4, App C4
FR-243	For a modular UMC, monitoring **SHALL** show participant progress across modules and periods.	G 11.3, 17.2
FR-244	The lecturer **SHALL** record the quality monitoring domains of Appendix C5 with status Satisfactory or Needs action and evidence.	App C5
FR-245	Lecturers (P1) and deputy deans (P2) **SHOULD** have a dashboard of their offerings showing indicators, open CQI actions, upcoming QV expiries and periodic reviews due.	G 17.2

3.15.2 CQI Records

ID	Requirement	Source
FR-246	The lecturer **SHALL** record issues or gaps with evidence, root cause, impact and priority (High, Medium, Low).	App C6
FR-247	The lecturer **SHALL** record an action plan per issue: improvement action, owner, due date, success indicator and status (Not started, In progress, Done).	App C7
FR-248	Each completed action **SHALL** be evaluated for effectiveness (Effective, Partly effective, Not effective) with before and after evidence. An action **SHALL** be closed only after this evaluation; an action that is not effective **SHALL** be carried to the next cycle.	G 17.12, App C8, BR-41
FR-249	The system **SHALL** link each CQI record along the chain issue, evidence, analysis, action, implementation and effectiveness check.	G 17.6, 18.9

ID	Requirement	Source
FR-250	A CQI action that affects learning outcomes, SLT or credit, MQF level or QV, core content, major assessment, delivery mode, credential or partner **SHALL** create a change request under Section 3.3.3.	G 17.7, App C9
FR-251	Each CQI record **SHALL** be prepared by the lecturer and confirmed by the deputy dean, covering the confirmations of Appendix C10.	App C10

3.15.3 Periodic Review, Suspension and Termination

ID	Requirement	Source
FR-252	Each micro-credential **SHALL** have a periodic review date. The system **SHALL** remind the lecturer and PTj before it is due.	G 11.5, 17.9
FR-253	A periodic review **SHALL** record its outcome as in Appendix C9: continue without significant change; continue with minor CQI; content or assessment review needed; review and re-approval needed; or consider suspension or termination.	App C9
FR-254	Suspension or termination of an offering **SHALL** be decided by the configured authority. Before the decision is final, the system **SHALL** list active participants and require a transition plan (completion window, mapping to a new version, or equivalence of earlier achievement).	G 17.10, App H7
FR-255	Affected participants **SHALL** be notified of the status of the offering, their transition options, the period to act and the implications for credentials or academic recognition.	G 17.10
FR-256	Suspension or termination **SHALL NOT** cancel valid achievements, credentials or ACR. All related records **SHALL** be kept for their retention period.	G 18.13, BR-40

3.16 Records, Audit Trail and Reporting (M15)

Every decision that affects the status of an offering, a participant's achievement, a credential, a credit value or the use of credit must have a source record, a decision-maker and a date that can be traced [G 15.7]. Corrections never erase history [G 18.10]. MCMS must allow the whole chain from development to CQI to be traced for any participant [G 18.14].

ID	Requirement	Source
FR-257	The system **SHALL** write an append-only audit entry for every approval and decision, credential action, result change, ACR, deposit, transfer and redemption status change, role and setting change, correction, and record disposal. Each entry **SHALL** hold time, actor, actor's role, action, record affected, values before and after, reason and external reference.	G 15.7, 18.10, D-20, C6
FR-258	Authorised users **SHALL** be able to search and view the audit trail of a record within their scope.	App H6
FR-259	A correction to any record **SHALL** follow the Guide procedure: check against the valid source record; carry out by the party with authority; record the original record, reason, action, approver or executor and date. If it affects a credential, ACR, deposit or credit status, the participant **SHALL** be notified and the linked records **SHALL** be updated in a controlled way.	G 18.10, App F10, BR-38
FR-260	The system **SHALL** keep a discrepancy log for mismatches between achievement records, credentials, ACR and ABC information, each resolved through the correction procedure.	G 12.10, 18.10
FR-261	For any participant and micro-credential the system **SHALL** show the traceability chain: development, approval, offering, learning, assessment, achievement, credential, ACR and ABC (if any), transfer and use of credit (if any), and monitoring and CQI.	G 18.14

ID	Requirement	Source
FR-262	For each cohort the system **SHALL** show the minimum records of Appendix H5 and whether each is present.	App H5
FR-263	The system **SHALL** provide standard reports, filtered by the user's scope: offerings by category, PTj and status; enrolments and completions; credentials issued, corrected and revoked; QV status and upcoming expiries; ACR prepared and deposits recorded; transfer applications and decisions; credit redemption status; open CQI actions.	G 17.2
FR-264	Reports **SHALL** be exportable to CSV, Excel and PDF.	
FR-265	Each record category **SHALL** have a retention period setting. Records **SHALL** be disposed of only after the period has passed and when not held for audit, quality assurance, academic recognition, legal reasons or participant protection. Each disposal **SHALL** be approved and logged.	G 18.12, A-08, OI-07
FR-266	An authorised user **SHOULD** be able to place a hold on records that stops disposal.	G 18.12
FR-267	The system **MAY** provide analytical dashboards across years, such as trends in enrolment, completion and credit use.	

3.17 Notifications and Integration Hub (M16)

ID	Requirement	Source
FR-268	The system **SHALL** send in-system notifications and email for at least these events: proposal submitted, reviewer assigned, review due, amendment requested, decision made; offering open; registration, payment and enrolment confirmed; results verified; credential issued, corrected, replaced or revoked; QV expiry alerts; consent requested; deposit recorded; transfer decision; credit expiry alert; CQI action due; periodic review due; change affecting enrolled participants.	G 10.3.2, 18.10
FR-269	Notifications **SHALL** be sent in the recipient's chosen language using the managed templates.	D-02
FR-270	Users **SHALL** be able to switch off optional notifications. Mandatory notices (decisions, credential actions, consent requests, changes affecting enrolment) **SHALL NOT** be switched off.	
FR-271	The system **SHALL** log each notification with recipient, channel, time and delivery status.	
FR-272	Approvers and lecturers **SHOULD** be able to receive a daily digest instead of individual emails.	
FR-273	The integration hub **SHALL** register each adapter (SSO, learning platform, ABC, QVC/MQA, payment, email, student information system) with its current mode: stub, manual or live.	D-06, D-14
FR-274	Failed outbound calls **SHALL** be retried with increasing delay. Calls that still fail **SHALL** be listed for P5 to resolve, without blocking the business process that created them.	

4. Business Rules

This section lists the rules of the Guide that MCMS enforces, as a separate requirement category (D-25). A business rule is binding on the complete system under UTeM policy, whatever the release (D-24). The column **Enforced by** names the functional requirements that put the rule into effect; acceptance tests SHALL cover every rule. Rules BR-01 to BR-06 keep the numbers given in the Glossary and Decision Log; their values are settings (Table 7).

4.1 Numeric Rules (Configurable)

ID	Rule	Source	Enforced by
BR-01	A SAMC and a whole UMC **SHALL** have at least 40 hours SLT (a UMC: or at least 1 credit).	G 6.3.1, 6.3.2	FR-31, FR-55
BR-02	Quality Verification **SHALL** be valid for at most 3 continuous years.	G 11.2.2	FR-174
BR-03	A SAMC **SHALL** have at least one completed cohort before it is submitted for QV.	G 11.2.1	FR-167
BR-04	Credit in ABC **SHALL** be valid for 10 years from the deposit date.	G 12.8	FR-196
BR-05	At least 30% of graduating credit for a stackable degree **SHALL** be earned directly at UTeM.	G 14.5.1	FR-232
BR-06	A participant graduating through a stackable pathway in a conventional programme **SHALL** complete at least 1 semester of residency at UTeM.	G 14.5.3	FR-233

4.2 Category and Design Rules

ID	Rule	Source	Enforced by
BR-07	An MCE has less than 40 hours SLT, carries no credit, is not subject to QV, and cannot be used for credit transfer or stackable pathways. It becomes credit-eligible only by reclassification and re-approval as a SAMC.	G 6.3.3	FR-30, FR-34
BR-08	The total SLT of all modules of a UMC **SHALL** equal the SLT of the parent course.	G 6.3.1, 8.3	FR-54
BR-09	A UMC **SHALL** come from a credit course of an accredited UTeM programme and **SHALL** keep its link to the parent course, with its outcomes mapped to the parent CLOs.	G 6.3.1, 8.2	FR-52, FR-59
BR-10	Under the credit-based unbundling model, every module **SHALL** have at least 40 hours SLT.	G 6.3.1.1(a)	FR-56
BR-11	Under the topic-based unbundling model, module SLT may vary to match topical learning clusters, modules need not reach 40 hours individually, and total module SLT **SHALL** equal parent course SLT.	G 6.3.1.1(b), Rajah 6	FR-57
BR-12	Under the separate learning and assessment model, content delivery **SHALL** be partitioned across learning modules while integrated summative assessment **SHALL** be isolated in a dedicated assessment module.	G 6.3.1.1(c), 8.6	FR-58
BR-13	Module completion or a Digital Badge **SHALL NOT** count as completion of the whole UMC or as academic credit.	G 8.7.1, 12.3.1	FR-141, FR-133
BR-14	Badges or modules from different UMCs or parent courses **SHALL NOT** form a UMC without an equivalence decision.	G 8.7.2, 12.2	FR-134

ID	Rule	Source	Enforced by
BR-15	A UMC module **SHALL NOT** be a credit unit unless specifically approved as a separately recognisable credit unit.	G 13.5, 14.2.1, D-18	FR-60, FR-208
BR-16	A SAMC **SHALL** have a credit value and MQF level only after Quality Verification, using the values in the QV decision.	G 6.3.2, 12.3.2	FR-43, FR-175
BR-17	Reaching 40 hours SLT **SHALL NOT** by itself give credit value or MQF level.	G 6.5, 11.2.1	FR-31, FR-167

4.3 Approval, Delivery and Credential Rules

ID	Rule	Source	Enforced by
BR-18	Approval to offer, Quality Verification and credit transfer **SHALL** be separate decisions; none implies another.	G 11.2, 15.4	FR-72, FR-178
BR-19	No micro-credential **SHALL** be offered before it is approved and its readiness is confirmed.	G 9.6	FR-86, FR-89
BR-20	A material change **SHALL** be reviewed and re-approved before it is implemented.	G 9.5, 17.7	FR-78
BR-21	Recognition **SHALL** be based on evidence of achievement, not on attendance or participation alone.	G 6.1, 8.6	FR-123
BR-22	Achievement **SHALL** be verified before a Digital Badge, Certificate or Digital Credential is issued.	G 10.4	FR-125, FR-116
BR-23	A Certificate or Digital Credential for the whole micro-credential **SHALL** be issued only after all completion requirements are met.	G 8.7.3, 10.5	FR-139, FR-142
BR-24	No credential or offering information **SHALL** suggest a credit value, MQF level or QV status that has not been formally obtained.	G 10.5, App E5	FR-152, FR-153
BR-25	When QV expires, is suspended or is withdrawn, UTeM **SHALL** stop claiming Quality-Verified status for the offering.	G 11.2.2	FR-177, FR-91

4.4 Credit, Transfer and Stackable Rules

ID	Rule	Source	Enforced by
BR-26	An ACR **SHALL** be prepared only for an eligible completed UMC or a completed Quality-Verified SAMC.	G 12.1.1, 12.4.1	FR-180
BR-27	The internal Achievement Record **SHALL** be kept distinct from the ACR.	G 8.7.5, 18.3	FR-132, FR-184
BR-28	An ABC deposit is storage of a credit record and **SHALL NOT** be treated as approval of credit transfer.	G 12.5, 18.6	FR-193
BR-29	An ACR **SHALL** be submitted to ABC only with the participant's consent.	G 12.4.2, 16.8	FR-188
BR-30	Expiry of credit validity **SHALL NOT** delete the ACR; the ACR is kept in perpetuity unless the owner or a recognised authority directs otherwise.	G 12.8	FR-198, FR-200
BR-31	Credit used toward one qualification **SHALL NOT** be used toward another: it is provisionally redeemed on transfer, redeemed on programme completion, and may become available again if the programme is not completed while the credit is valid.	G 12.9	FR-216, FR-219
BR-32	Academic equivalence **SHALL** be judged on learning outcomes and related criteria, not on equal credit or SLT alone.	G 13.3	FR-206

ID	Rule	Source	Enforced by
BR-33	Acceptance of credit **SHALL** be decided by the receiving programme or institution; neither ABC nor MCMS decides it.	G 12.5, 13.1	FR-210
BR-34	An MCE **SHALL NOT** be used directly for credit transfer.	G 13.2	FR-202
BR-35	A refused transfer **SHALL NOT** cancel a valid ACR, badge, certificate or Digital Credential.	G 13.6	FR-211
BR-36	Programme admission **SHALL** be decided separately from credit transfer; holding micro-credentials, ACR or ABC credit **SHALL NOT** give automatic admission.	G 14.4	FR-214, FR-236
BR-37	Only credit-bearing credentials **SHALL** count toward a stackable degree.	G 14.1.1	FR-226
BR-38	Credit used for a stackable degree **SHALL** be mapped to the programme learning outcomes and Body of Knowledge.	G 14.5.2	FR-229
BR-39	A qualification **SHALL** be awarded only after lawful admission and completion of all programme and graduation requirements.	G 14.10	FR-237

4.5 Governance and Record Rules

ID	Rule	Source	Enforced by
BR-40	Corrections **SHALL** be made only by the party with authority and **SHALL** keep an audit trail; earlier actions are never erased.	G 10.5.1, 15.7, 18.10	FR-164, FR-259, FR-257
BR-41	A version of a micro-credential linked to participant records **SHALL NOT** be deleted while its records must be kept.	G 18.8	FR-81
BR-42	Suspension or termination of an offering **SHALL** protect current participants and **SHALL NOT** cancel valid achievements.	G 17.10, 18.13	FR-254, FR-256
BR-43	A CQI action **SHALL** be considered complete only after its effectiveness is evaluated.	G 17.12	FR-248
BR-44	Lecturers and operational units **SHALL NOT** take academic decisions outside the authority given to them.	G 15.2.3, 16.3	FR-15, FR-16

5. External Interface Requirements

5.1 User Interfaces

MCMS has three web front ends sharing one design: the **staff portal** (lecturers, deputy deans, CODL administrators and invited experts), the **participant portal** (P8) and the **public pages** (catalogue and credential verification). Figure 17 shows the main screens by audience. Detailed layouts belong to the SDD and the design work.

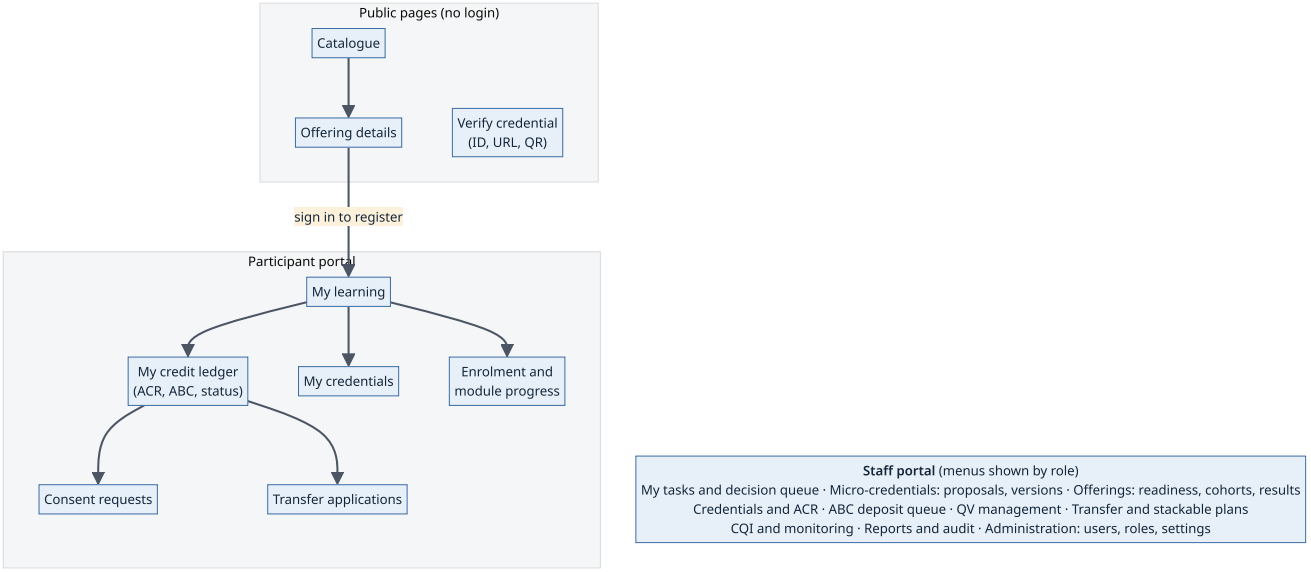


Figure 17: *Main screens by audience. Each staff user sees only the menus their roles allow.*

ID	Requirement	Source
IR-01	The user interface **SHALL** be a responsive web interface usable on screens from 360 px wide (phones) to desktop monitors.	
IR-02	Every screen **SHALL** be available in Bahasa Melayu and English, with a language switch on every page.	D-02
IR-03	After sign-in each staff user **SHALL** land on a task page that lists the items waiting for their action, across all their roles.	
IR-04	Each recognition stage (approval, badge, certificate or Digital Credential, QV, ACR, ABC deposit, credit transfer, redemption) **SHALL** have a distinct label and visual style, used consistently, so that users cannot confuse one stage with another.	D-15
IR-05	Forms **SHALL** validate input on entry, show errors next to the field in plain language, and keep entered data when an error occurs.	
IR-06	Actions that issue, approve, revoke, deposit or redeem **SHALL** ask for confirmation and show a summary of their effect before they run.	G 15.7
IR-07	Proposal documents, decision packs, certificates, ACR views and reports **SHALL** have print-friendly layouts.	App A, E3
IR-08	The interface and credentials **SHALL** follow UTeM corporate identity guidelines, using DESIGN.md (Carbon with uLearn Navy) and displaying the official Centre for Open and Distance Learning (CODL) logo in the public website footer (D-33, D-44, A-10).	App E3

ID	Requirement	Source
IR-09	The interface across all roles **SHALL** distinguish between operational status indicators (such as `Passed (Final)`, `0 / 2 eligible`, `Provisional (1/3)`, `Revoked`, and `Replaced`) and technical implementation details. Technical cryptographic labels (e.g. `Ed25519`, `W3C VC 2.0`, `RFC 7519`) and redundant status badges beside completed sections **SHALL NOT** be rendered as decorative pills on user-facing screens.	D-42
IR-10	Each screen **SHALL** have context help, and the help centre **SHALL** explain the separate recognition stages in plain words.	G 10.3.1
IR-11	The interface **SHALL** meet WCAG 2.1 level AA (see Section 7.4).	

5.2 Hardware Interfaces

MCMS has no special hardware interface. It runs on standard UTeM server infrastructure.

ID	Requirement	Source
IR-12	Verification QR codes **SHALL** be readable by the standard camera application of current smartphones and **SHALL** open the verification page directly.	App E6
IR-13	Certificates **SHALL** print correctly on standard A4 office printers.	
IR-14	Credential signing keys **MAY** be held in a hardware security module or an equivalent managed key store.	OI-08

5.3 Software Interfaces

All external systems are reached through adapters. Each adapter has a fixed interface towards the MCMS modules and one or more implementations: **stub** (simulated, for development and testing), **manual** (staff exchange files or enter data), and **live** (automated connection).

Table 10: *Adapters and their modes by release*

Adapter	External system	Data exchanged	Release 1	Later
Identity	UTeM SSO (Microsoft education account)	Sign-in, name, email, staff or matric number, account attributes	Live in production; stub in development and test	–
Learning platform	UCreds	Enrolments out; progress, results, module completion, badge data in	Manual (CSV)	Live connector (R2)
ABC	National Academic Bank of Credit	ACR deposit data out; deposit reference and status in; redemption updates	Manual (export and record)	Live (R3, **MAY**)
Quality Verification	QVC / MQA	QV submission package out; QV decision in	Manual (package and record)	Electronic (R3, **MAY**)
Payment	UTeM payment hub	Payment request out; payment result in	Stub (dummy service)	Live (R3, **MAY**)
Email	UTeM SMTP service	Notification messages	Live	–
Student information	UTeM student information system	Admission, enrolment and graduation status	None (manual recording)	Live (R3, **MAY**)
Credential signing	Key store within UTeM infrastructure	Signing of Digital Credentials; public key for verification	Live	–

ID	Requirement	Source
IR-15	Each adapter **SHALL** expose a documented, versioned interface to the MCMS modules. Business modules **SHALL NOT** call external systems directly.	D-06, C3
IR-16	Every adapter **SHALL** have a stub implementation usable in development and test environments.	D-14
IR-17	The active implementation of each adapter **SHALL** be chosen by configuration, without code change.	D-14
IR-18	Operations that send data out (deposit export, payment request, notification) **SHALL** be safe to repeat: a repeated call **SHALL NOT** create a duplicate record.	
IR-19	The identity adapter **SHALL** use OpenID Connect with the UTeM Microsoft tenant, unless CODL specifies another protocol (TBD, OI-03).	D-10
IR-20	CSV templates for the learning platform **SHALL** be versioned and published in the technical documentation, with UTF-8 encoding, a header row and ISO 8601 dates.	A-04
IR-21	Digital Credentials **SHALL** be produced as JSON documents following the Open Badges 3.0 / W3C Verifiable Credentials data model, signed by UTeM, with the public verification key published at a stable address.	D-19
IR-22	The system **SHOULD** provide an authenticated REST API for read access to catalogue data and credential verification, to allow later integration by other UTeM systems.	

5.4 Communications Interfaces

ID	Requirement	Source
IR-23	All web traffic **SHALL** use HTTPS with TLS 1.2 or later. Plain HTTP requests **SHALL** be redirected to HTTPS.	OWASP ASVS
IR-24	Email **SHALL** be sent through the UTeM SMTP service using TLS, from a monitored UTeM address.	
IR-25	Servers **SHALL** synchronise time with a trusted time source. Times **SHALL** be stored in UTC and shown in Malaysia time (Asia/Kuala_Lumpur).	G 15.7
IR-26	Dates **SHALL** be shown as DD/MM/YYYY in the interface and exchanged as ISO 8601 in files and APIs.	
IR-27	Files exchanged with external parties **SHALL** be transferred only through approved UTeM channels and **SHALL NOT** be sent by personal email.	App H6

6. Data Requirements

This section describes the information MCMS must hold, as a conceptual model. It names the main entities and their relationships so that reviewers can check that every record required by the Guide exists, and so that the SDD can derive the physical database design. Table names, column types and indexes belong to the SDD.

6.1 Conceptual Data Model

The model is shown in three parts. The most important property is visible in all three: each recognition stage is its own entity (approval decision, Digital Badge, credential, QV record, ACR, ABC deposit, transfer decision, redemption event), never a status field on another entity (D-15).

6.1.1 Curriculum and Approval

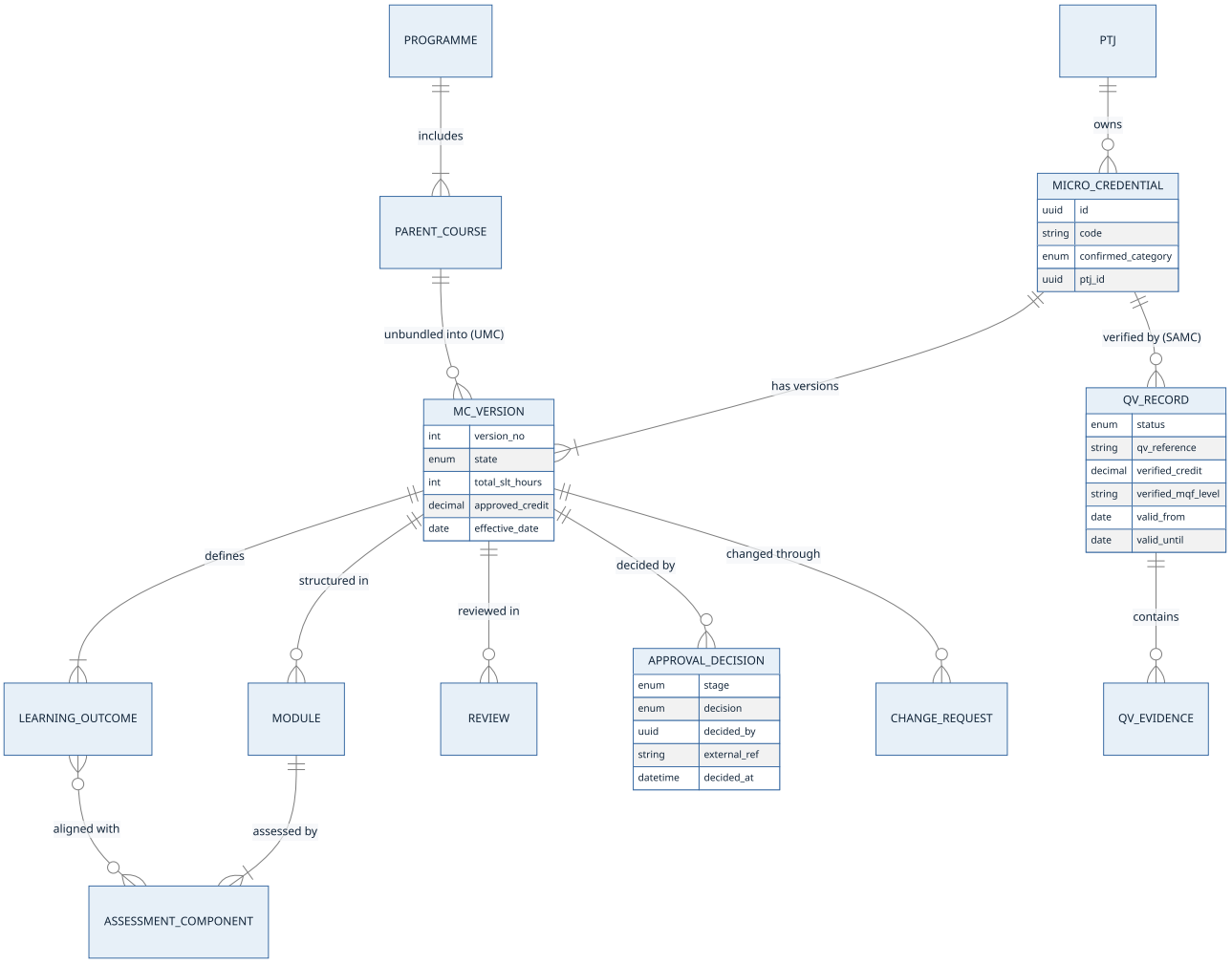


Figure 18: *Conceptual model of curriculum, versions, approval and Quality Verification. Only a few key attributes are shown.*

6.1.2 Delivery, Achievement and Credentials

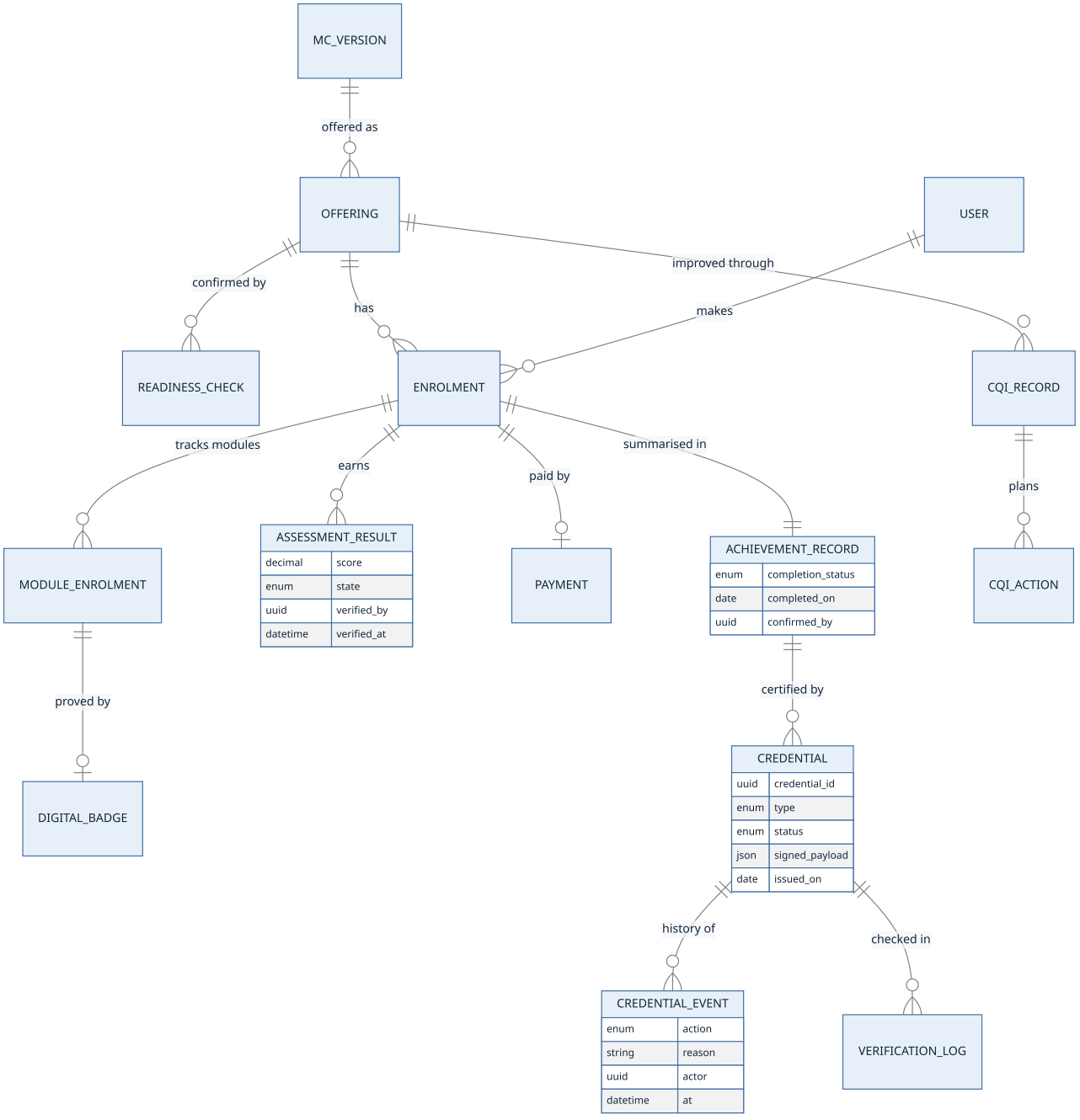


Figure 19: *Conceptual model of offerings, enrolment, results, the Achievement Record and credentials.*

6.1.3 Credit, Transfer and Stackable Pathway

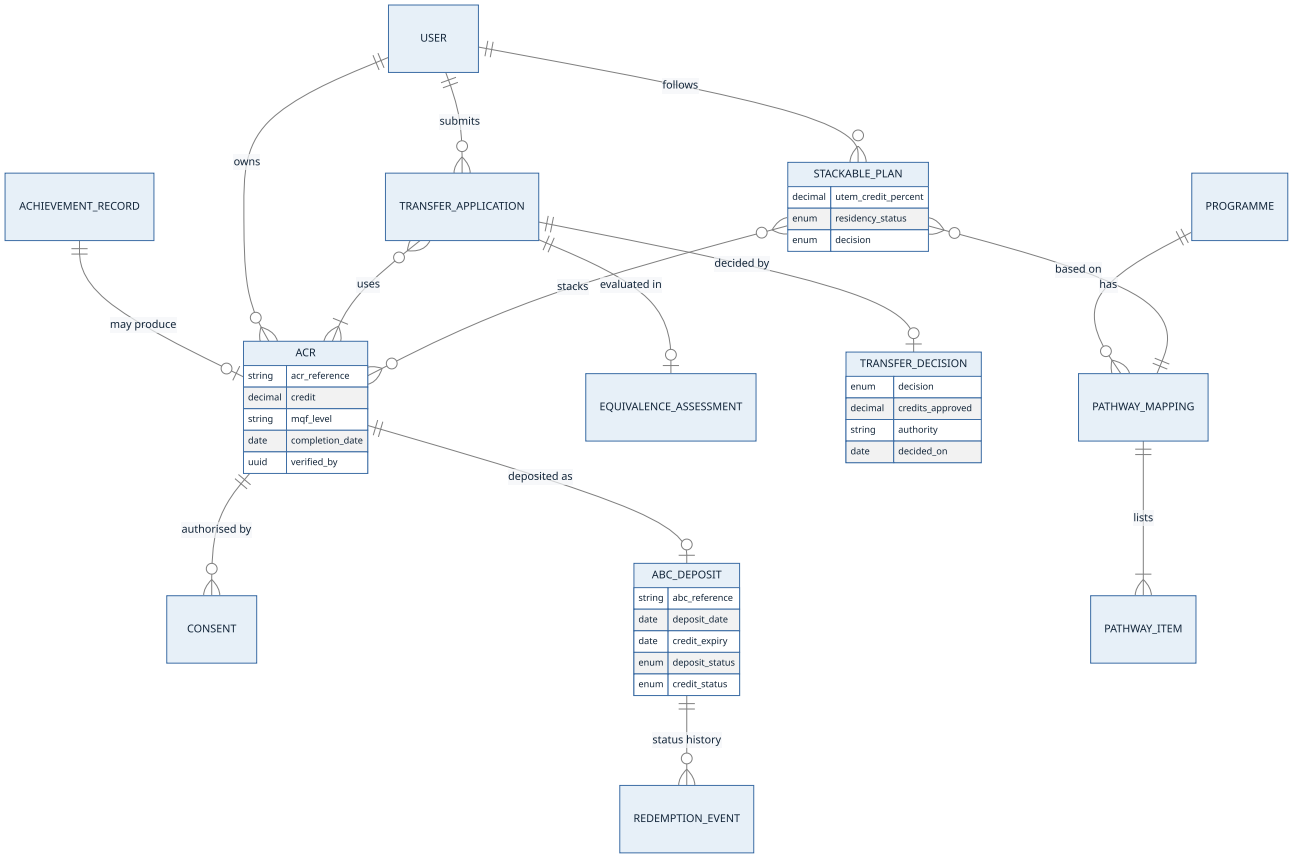


Figure 20: *Conceptual model of ACR, ABC deposit, redemption, credit transfer and stackable plans.*

Table 11: *Main entities and their purpose*

Entity	Purpose	Owner (App H4)	Main requirements
PTJ, PROGRAMME, PARENT_COURSE	Reference data for owners, target programmes and parent courses with accreditation status, credits, SLT and CLOs	S1	FR-25
MICRO_CREDENTIAL, MC_VERSION	A micro-credential and its immutable approved versions	P1 / P2	FR-76
LEARNING_OUTCOME, MODULE, ASSESSMENT_COMPONENT	Learning design and alignment	P1	FR-38, FR-53
REVIEW, APPROVAL_DECISION, CHANGE_REQUEST	Review results, decisions per stage, minor and material changes	P1 / P2 (records Senate decisions)	FR-69, FR-77
QV_RECORD, QV_EVIDENCE	QV cycles, evidence checklist, verified credit and MQF, validity	P1 / P2	FR-174
OFFERING, READINESS_CHECK	Cohorts and their readiness confirmation	P2	FR-85, FR-88
USER, ROLE_ASSIGNMENT	People, roles and scopes	S1	FR-13

Entity	Purpose	Owner (App H4)	Main requirements
ENROLMENT, MODULE_ENROLMENT, PAYMENT	Registration, module progress and fee status	PTj (P2); CODL operates the data	FR-105, FR-106
ASSESSMENT_RESULT	Provisional and verified results	P1 / P2	FR-125
ACHIEVEMENT_RECORD	Internal record of modules, badges and completion; not an ACR	P1 / P2	FR-131
DIGITAL_BADGE, CREDENTIAL, CREDENTIAL_EVENT	Proofs of achievement and their history	P2 (issued by the system)	FR-143, FR-161
ACR, CONSENT, ABC_DEPOSIT, REDEMPTION_EVENT	Academic credit records, consent, deposit and credit status history	P2 (ACR generated by the system)	FR-182, FR-192, FR-216
TRANSFER_APPLICATION, EQUIVALENCE_ASSESSMENT, TRANSFER_DECISION	Credit transfer cases	P2 of the degree faculty	FR-201, FR-209
PATHWAY_MAPPING, PATHWAY_ITEM, STACKABLE_PLAN	Programme pathway mappings and individual plans	P1 / P2; Senate approval recorded by P2	FR-225, FR-231
CQI_RECORD, CQI_ACTION	Monitoring findings and improvement actions	P1 / P2	FR-247
AUDIT_LOG, SETTING, NOTIFICATION, INTEGRATION_JOB	Audit trail, configuration, messages and adapter jobs	System	FR-257, FR-21

6.2 Data Integrity Rules

ID	Requirement	Source
DR-01	Each recognition stage **SHALL** be stored in its own entity with its own decision fields. No single status field **SHALL** represent more than one stage.	D-15
DR-02	Offerings, enrolments, results, credentials, ACR and equivalence evaluations **SHALL** reference the exact MC version they belong to.	G 13.8, 18.8
DR-03	Approved versions, issued credential payloads, verified results, decisions and audit entries **SHALL NOT** be updated in place. A change **SHALL** create a new record linked to the previous one.	D-20, G 18.10
DR-04	Records **SHALL NOT** be physically deleted except by the approved disposal process after their retention period.	G 18.12
DR-05	Credit value, MQF level and QV status shown anywhere **SHALL** be derived from the approval or QV record, with a reference to that record.	D-17
DR-06	Each decision **SHALL** store the settings values it used (for example the credit validity period), so that later setting changes do not alter past decisions.	D-16
DR-07	Records exposed outside the staff portal (credentials, verification links) **SHALL** use random, non-sequential identifiers.	
DR-08	All timestamps **SHALL** be stored in UTC with time zone.	
DR-09	Relationships **SHALL** be enforced by the database (foreign keys and constraints), not only by application code.	D-09

ID	Requirement	Source
DR-10	Text that participants see (titles, outcomes, recognition statements) **SHALL** be stored in both Bahasa Melayu and English.	D-02

6.3 Data Retention and Ownership

The Guide requires retention according to UTeM's records policy and, for ACR and ABC data, the ABC rules [G 18.12]. The periods below are placeholders (A-08, OI-07); only the ACR rule is fixed by the Guide.

Table 12: *Record categories, owners and retention (placeholders)*

Record category	Owner / custodian	Retention	Source
Development and approval documents, versions	PTj / designated party	TBD (UTeM records policy); versions linked to participants kept while those records are kept	G 18.2, 18.8
Learning materials and versions	PTj / academic owner	TBD	App H4
Assessment instruments, rubrics and results	PTj / academic party	TBD	App H4
Registration records	PTj (P2)	TBD	App H4
Achievement Record	PTj (P2)	TBD	G 18.3
Badges, Certificates, Digital Credentials and their history	PTj (P2); issued by the system	TBD; verification remains possible for the life of the credential	App E6, H4
QA, monitoring and CQI records	PTj (P2)	TBD	G 18.9
QV documents and decisions (SAMC)	PTj (P2)	At least the QV validity period plus records requirement	App H4
Academic Credit Record (ACR)	PTj (P2); generated by the system	Indefinite, unless the owner or a recognised authority directs otherwise	G 12.8
ABC deposit records	PTj (P2)	Per ABC rules	App H4
Credit transfer decisions	Academic / records party	TBD (academic regulations)	App H4
Stackable mapping matrices	Programme / PTj	Programme lifetime plus review period	App H4
Audit trail	System (CODL)	At least as long as the longest linked record	G 15.7
Verification request logs	System (CODL)	TBD	App E6

6.4 Data Migration and Reference Data

ID	Requirement	Source
DR-11	The system **SHALL** import reference data for PTj, programmes and parent courses from CSV files supplied by the responsible offices.	
DR-12	The system **SHOULD** import existing micro-credential records kept in spreadsheets (offerings, participants, completions) as historical records, marked as migrated and not re-verified.	

ID	Requirement	Source
DR-13	Migrated records **SHALL NOT** produce ACR or credit claims unless their eligibility is verified under Section 3.11.	G 12.1, D-17

6.5 Data Classification

Table 13: *Data classification*

Class	Examples	Handling
Public	Catalogue entries, public verification fields, issuer public key	May be shown without login
Internal	Proposals, reviews, decisions, CQI records, reports	Staff within scope only
Confidential	Participant identity (name, identity card or passport number, matric number, contact details), results, ACR, consent, transfer cases	Role and scope controlled; encrypted at rest; minimum exposure
Restricted	Credential signing keys, system secrets, audit integrity data	Administrators of the key store only; never shown in the interface

7. Non-Functional Requirements

This section states the quality the system must achieve. Figures for load are based on the indicative sizing of A-09 (up to 200 offerings and 20,000 participants a year, peaks of 2,000 registrations within a few hours); they are not binding commitments (D-21) and will be revised when real figures are known.

7.1 Performance Requirements

7.1.1 Response Time

ID	Requirement	Source
NFR-01	95% of page requests **SHALL** complete within 2 seconds under normal load, measured at the server.	
NFR-02	Catalogue and record searches **SHALL** return the first page of results within 2 seconds for 95% of requests.	
NFR-03	The public verification page **SHALL** respond within 1 second for 95% of requests.	App E6
NFR-04	Standard reports for one PTj and one year **SHALL** be produced within 10 seconds; larger reports **SHALL** run in the background and notify the user when ready.	

7.1.2 Throughput

ID	Requirement	Source
NFR-05	The system **SHALL** handle 2,000 registrations within 2 hours, with at least 100 concurrent users submitting, without errors or loss of data.	A-09
NFR-06	Registration **SHALL** never accept more participants than the capacity of an offering, even under concurrent submissions.	
NFR-07	A CSV import of 5,000 result rows **SHALL** be validated and staged within 2 minutes.	
NFR-08	Issuing 500 signed Digital Credentials for a cohort **SHALL** complete within 10 minutes as a background job.	

7.1.3 Resource Utilisation and Scalability

ID	Requirement	Source
NFR-09	At the peak load of NFR-05, average CPU use of application servers **SHALL** stay below 70%, leaving headroom for unexpected load.	D-21
NFR-10	Application servers **SHALL** keep no session state on local disk, so that more servers can be added behind a load balancer.	D-21
NFR-11	Long-running work (imports, credential issuance, reports, notifications, alerts) **SHALL** run in background queues, so that interactive response times are not affected.	
NFR-12	The design **SHALL** allow ten times the indicative annual volume without a change of architecture.	D-21

7.2 Security Requirements

7.2.1 Authentication and Authorisation

ID	Requirement	Source
SR-01	In production, all users except public verifiers **SHALL** authenticate through UTeM SSO. MCMS **SHALL NOT** store user passwords.	D-10

ID	Requirement	Source
SR-02	Staff roles that approve, verify, issue, deposit or administer **SHALL** require multi-factor authentication, enforced through the SSO provider where possible.	
SR-03	Every request **SHALL** be checked on the server against the user's roles and scope. Access to a record outside scope **SHALL** be refused and logged.	G 18.11
SR-04	Roles **SHALL** follow least privilege: each role **SHALL** have only the permissions needed for its functions in Table 6.	App H6
SR-05	S1 **SHALL** be able to produce a list of all role assignments for periodic access review.	App H6

7.2.2 Data Protection

ID	Requirement	Source
SR-06	Data in transit **SHALL** be encrypted (Section 5.4).	
SR-07	The database, file storage and backups **SHALL** be encrypted at rest. Identity card and passport numbers **SHALL** additionally be encrypted at field level.	PDPA
SR-08	Credential signing keys **SHALL** be generated and stored in a protected key store, **SHALL** never leave it in readable form, and **SHALL** be rotatable without invalidating credentials already issued. Key ownership is TBD (OI-08).	D-19
SR-09	Uploaded files **SHALL** be checked for allowed type and size, scanned for malware where a scanner is available, and stored outside the public web root.	
SR-10	The audit trail **SHALL** be tamper-evident (for example, each entry carries a hash of the previous one), and the system **SHALL** be able to check its integrity.	D-20
SR-11	Backups **SHALL** be stored separately from the production system with access restricted to authorised administrators.	App H6

7.2.3 Privacy and Compliance

ID	Requirement	Source
SR-12	Processing of personal data **SHALL** comply with PDPA 2010: notice at collection, use only for stated purposes, disclosure only as permitted, and retention only as long as needed.	PDPA, G 18.11
SR-13	The public verification page and any shared data **SHALL** show only the minimum permitted information (Section 3.9.3).	App E6, H6
SR-14	A participant **SHALL** be able to see the personal data MCMS holds about them and request correction through the correction procedure.	PDPA, G 18.10
SR-15	Data **SHALL** be shared with external parties (ABC, QVC/MQA, partners) only through approved mechanisms and only to the extent needed.	App H6

7.2.4 Application Security, Monitoring and Incident Response

ID	Requirement	Source
SR-16	The application **SHALL** meet OWASP ASVS 4.0 level 2, including input validation, output encoding, protection against cross-site request forgery and injection, and secure session cookies.	
SR-17	Public endpoints (catalogue, verification) **SHALL** be rate-limited to resist automated abuse and enumeration of credential IDs.	
SR-18	Security events (sign-in, failed authorisation, role changes, key use, export of personal data) **SHALL** be logged and **SHALL** be available to UTeM ICT security monitoring.	

ID	Requirement	Source
SR-19	The system **SHOULD** alert administrators to suspicious patterns, such as many failed authorisations or mass data export.	
SR-20	An incident procedure **SHALL** be documented, covering detection, containment, notification of affected participants where required, and recovery.	PDPA
SR-21	A security test **SHALL** be carried out before the first production release and after major changes.	

7.3 Reliability and Availability

ID	Requirement	Source
NFR-13	The system **SHALL** be available 99.5% of the time during UTeM working hours (08:00–18:00, Monday to Friday), excluding announced maintenance.	
NFR-14	The public verification page **SHOULD** be available 99.5% of the time at all hours, because employers may check credentials at any time.	App E6
NFR-15	Planned maintenance **SHALL** be announced to users at least 48 hours in advance and **SHALL** be scheduled outside working hours unless an urgent fix requires otherwise.	
NFR-16	The database and stored files **SHALL** be backed up daily, with backups kept for at least 30 days. The recovery point objective (RPO) **SHALL** be 24 hours.	App H6
NFR-17	The system **SHALL** be restorable from backup within a recovery time objective (RTO) of 8 working hours. A restore **SHALL** be tested at least twice a year.	
NFR-18	Failure or unavailability of an external system **SHALL NOT** stop the business functions that do not depend on it; affected jobs **SHALL** wait and retry.	D-06
NFR-19	Errors **SHALL** be shown to users as clear messages in their language with a reference code, without technical details. Full details **SHALL** be logged for support staff.	
NFR-20	Each business action that changes several records (for example approving a transfer and changing credit status) **SHALL** succeed or fail as a whole.	

7.4 Usability and Accessibility

ID	Requirement	Source
NFR-21	The interface **SHALL** conform to WCAG 2.1 level AA, including keyboard operation, sufficient colour contrast, text alternatives and labels readable by screen readers. Status **SHALL** never be shown by colour alone.	
NFR-22	Labels, messages and help **SHALL** use plain language. Each recognition status **SHALL** have a short explanation available on the screen where it appears.	G 10.3.1
NFR-23	A trained staff user **SHALL** be able to complete a common task (verify a cohort's results, issue credentials, record a deposit) in no more than five steps from their task page.	
NFR-24	Participant functions (catalogue, registration, progress, credentials, consent, ledger) **SHALL** be fully usable on a phone.	
NFR-25	Switching language **SHALL** keep the user on the same page with entered data intact.	D-02
NFR-26	Pages for participants **SHOULD** remain usable on slow mobile connections (for example, page weight under 1.5 MB without media).	

7.5 Maintainability and Portability

ID	Requirement	Source
NFR-27	The code **SHALL** be organised into the modules of Table 2, each with its own internal interface. A module **SHALL NOT** read or write another module's tables directly.	D-07
NFR-28	The system **SHALL** follow the conventions of the Laravel framework and the PHP-FIG coding standards, so that CODL PHP developers can maintain it.	D-08
NFR-29	Every business rule of Section 4 **SHALL** have at least one automated test. The automated test suite **SHALL** run on every change.	D-01, D-05
NFR-30	Database access **SHALL** use the Laravel database layer, keeping vendor-specific features to a documented minimum, so that the system can run on PostgreSQL or MySQL/MariaDB.	D-09
NFR-31	Environment-specific values (addresses, keys, adapter modes) **SHALL** be configuration, not code.	D-14
NFR-32	Database changes **SHALL** be applied through versioned migration scripts.	
NFR-33	The system **SHALL** be deployable to the development, test and production environments by a documented, repeatable procedure, preferably as containers.	
NFR-34	The SDD, adapter interfaces, CSV templates and operating procedures **SHALL** be kept up to date with each release.	

7.6 Legal and Compliance Requirements

ID	Requirement	Source
NFR-35	The system **SHALL** comply with the Guide and the MQA guidelines it refers to, as they apply to the controlling copy of the Guide (Versi 1.0, 18 September 2026).	D-03
NFR-36	The system **SHALL** comply with the Personal Data Protection Act 2010 and UTeM data protection policy (Section 7.2.3).	
NFR-37	The system **SHALL** support UTeM's records and document management policy, including retention and controlled disposal.	G 18.12
NFR-38	Software components used **SHALL** have licences compatible with UTeM ownership and on-premises use. The use of third-party learning materials is governed by Section 3.2.2.	G 9.3
NFR-39	Digital Credentials **SHALL** follow the open standards of reference 9 so that they can be verified by third parties without UTeM software.	D-19

7.7 Operational Requirements

ID	Requirement	Source
NFR-40	The system **SHALL** expose health checks and **SHALL** be monitored for availability, response time, queue length, failed jobs and disk use, with alerts to the operations team.	
NFR-41	Application logs **SHALL** be structured, time-stamped and kept for at least 90 days. Logs **SHALL NOT** contain identity card numbers, tokens or secrets.	PDPA
NFR-42	Routine administration (users, roles, settings, templates, reference data, adapter modes) **SHALL** be possible through the interface without database access.	D-16
NFR-43	Timed jobs (expiry alerts, overdue reminders, retention checks) **SHALL** run on a scheduler, and each run **SHALL** be logged with its result.	

ID	Requirement	Source
NFR-44	Development and test environments **SHALL NOT** contain real personal data unless it is masked.	PDPA

7.8 Cultural and Language Considerations

ID	Requirement	Source
NFR-45	Name fields **SHALL** accept long names and Malaysian naming forms (for example bin, binti, a/l, a/p) and **SHALL** print them in full on certificates.	App E3
NFR-46	The Bahasa Melayu interface **SHALL** use the terms of the Guide (for example Rekod Pencapaian Pembelajaran, Pemindahan Kredit, Penebusan Tunggal) consistently.	D-02
NFR-47	Reminders and due dates **SHOULD** take account of Malaysian public holidays and the UTeM academic calendar.	

8. Other Requirements

8.1 Internationalisation Requirements

ID	Requirement	Source
NFR-48	All interface text, notifications, templates and recognition statements **SHALL** be held in language resource files or managed templates, never fixed in code, so that a further language could be added.	D-02
NFR-49	Where a bilingual content field has only one language filled, the system **SHALL** show the available language with a marker rather than an empty field.	
NFR-50	Fees **SHALL** be shown in Malaysian ringgit (RM) with two decimal places.	

8.2 Training Requirements

ID	Requirement	Source
NFR-51	Role-based training **SHALL** be provided before go-live for lecturers, deputy deans and CODL administrators and technical staff, with a short guide for students.	D-28
NFR-52	Each role **SHALL** receive a quick reference card in Bahasa Melayu and English.	D-02
NFR-53	A training environment with sample data and the stub adapters **SHOULD** be available so that staff can practise without affecting real records.	D-14
NFR-54	CODL staff **SHOULD** be trained as trainers so that later staff can be trained internally.	

8.3 Acceptance Approach

A separate Test Plan is not part of the controlling document set; acceptance tests are traced directly from the requirement IDs of this SRS (D-01). For Release 1, acceptance requires that:

1. every SHALL requirement of Sections 3, 5, 6 and 7 has at least one passing acceptance test that names its ID;
2. every business rule of Section 4 has at least one passing automated test, including a test that the system refuses the forbidden case (for example, an ACR for an MCE, or reuse of redeemed credit);
3. an end-to-end walkthrough of Figure 25 is completed in the test environment with stub adapters;
4. open items that block a Release 1 requirement (Appendix C) are either resolved or covered by the documented working assumption.

Appendix A: Analysis Models

This appendix gathers models that show how the requirements work together. They are explanatory; the requirements in Sections 3–7 prevail.

A.1 Use Case Overview: Staff

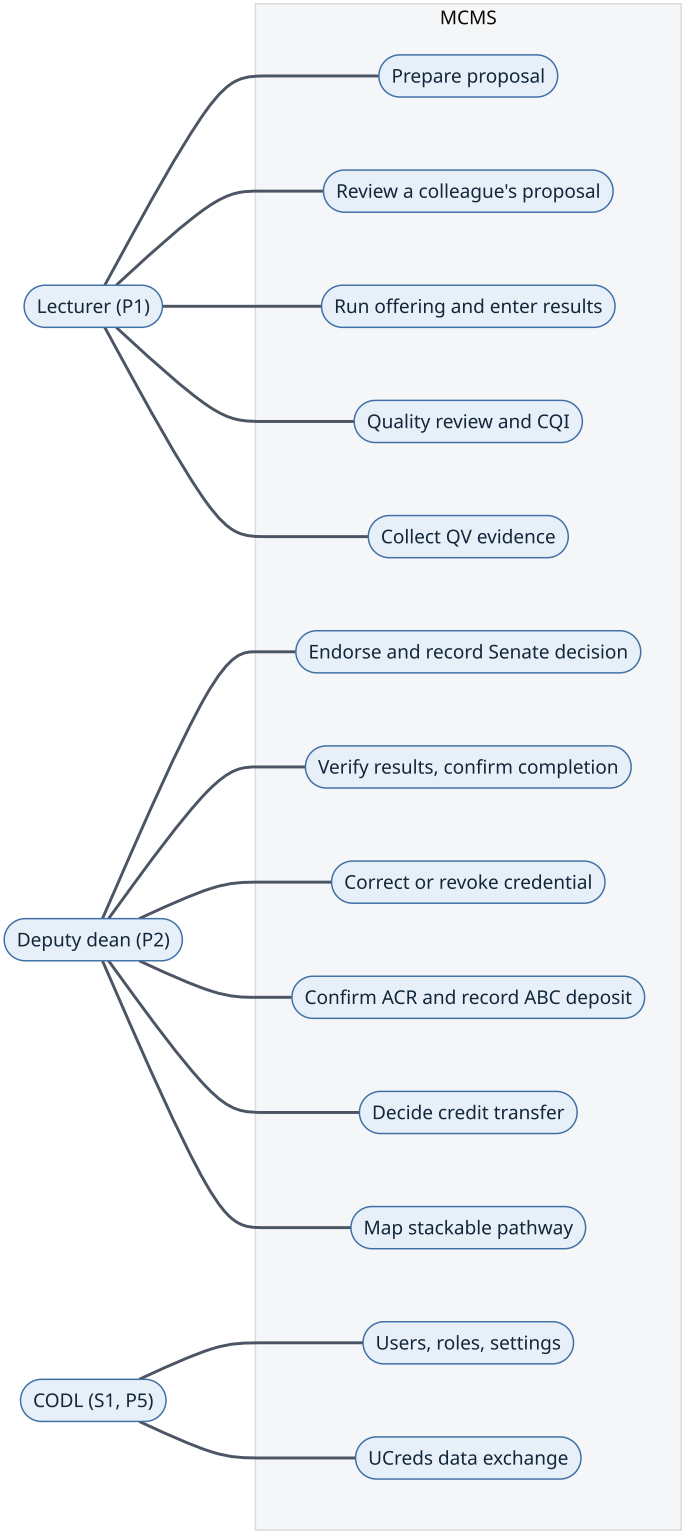


Figure 21: *Main use cases of staff users. When the deputy dean verifies results or confirms completion, the system issues credentials and generates ACR automatically (D-29). The Senate does not log in; its decisions are recorded by the deputy dean.*

A.2 Use Case Overview: Participants and Public

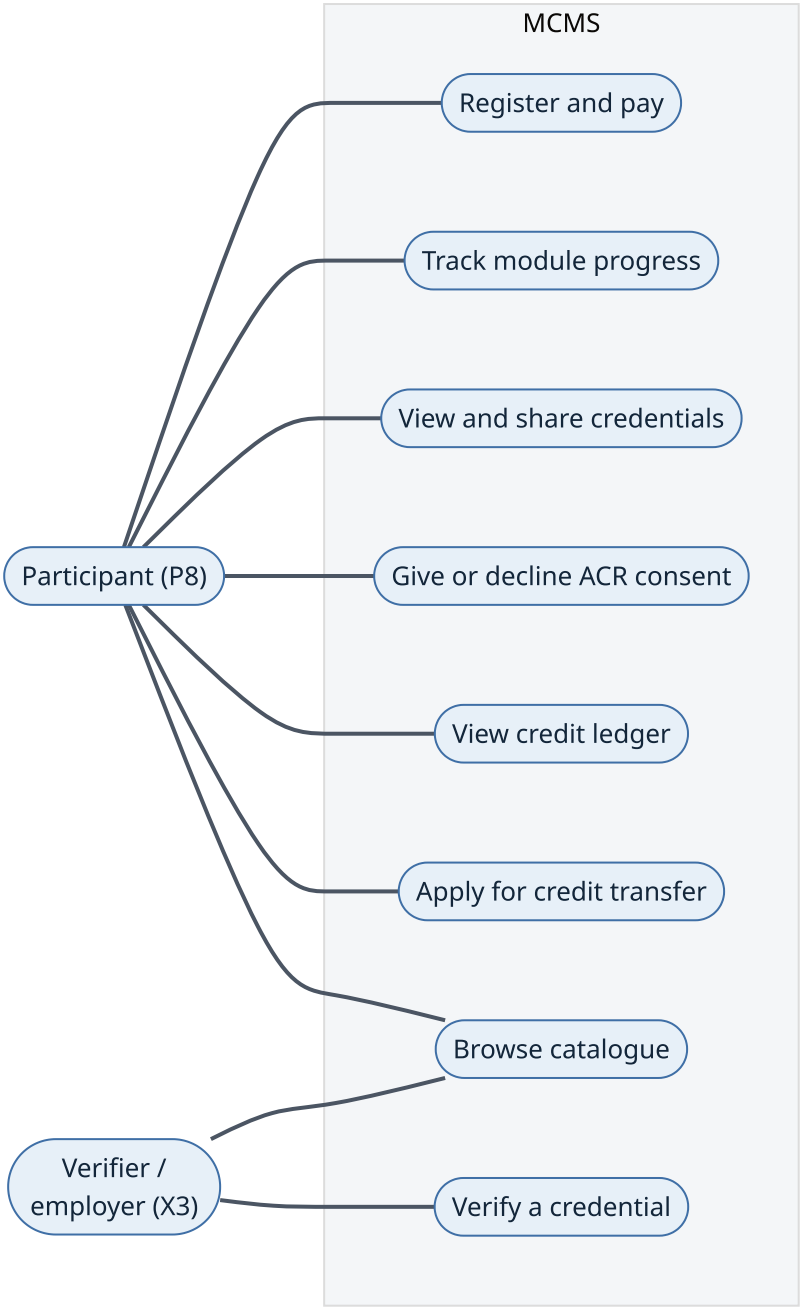


Figure 22: *Main use cases of participants and the public.*

A.3 Sequence: From Proposal to Approved Offering

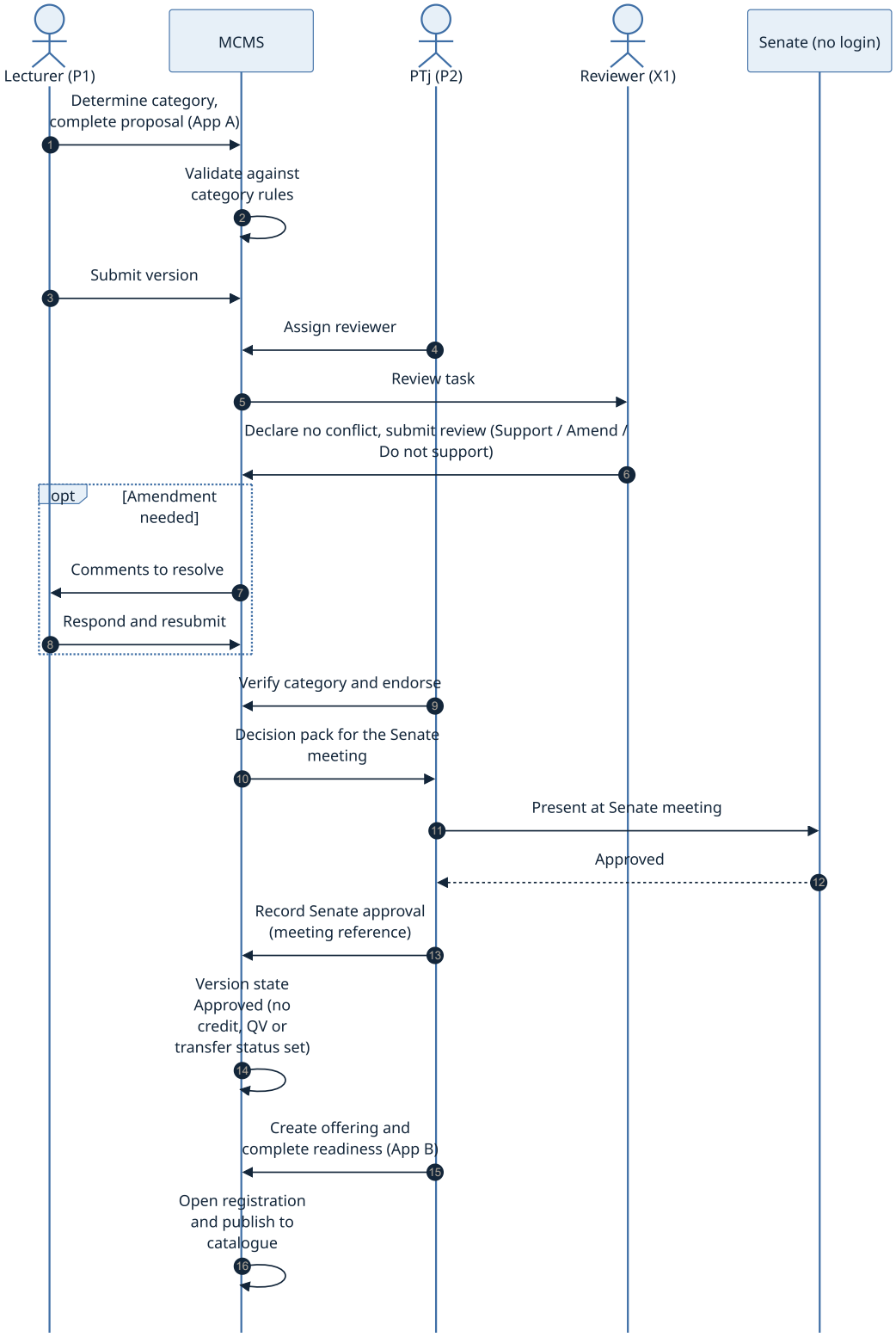


Figure 23: *Proposal, review, approval and offering.*

A.4 Sequence: Credential Issuance and Public Verification

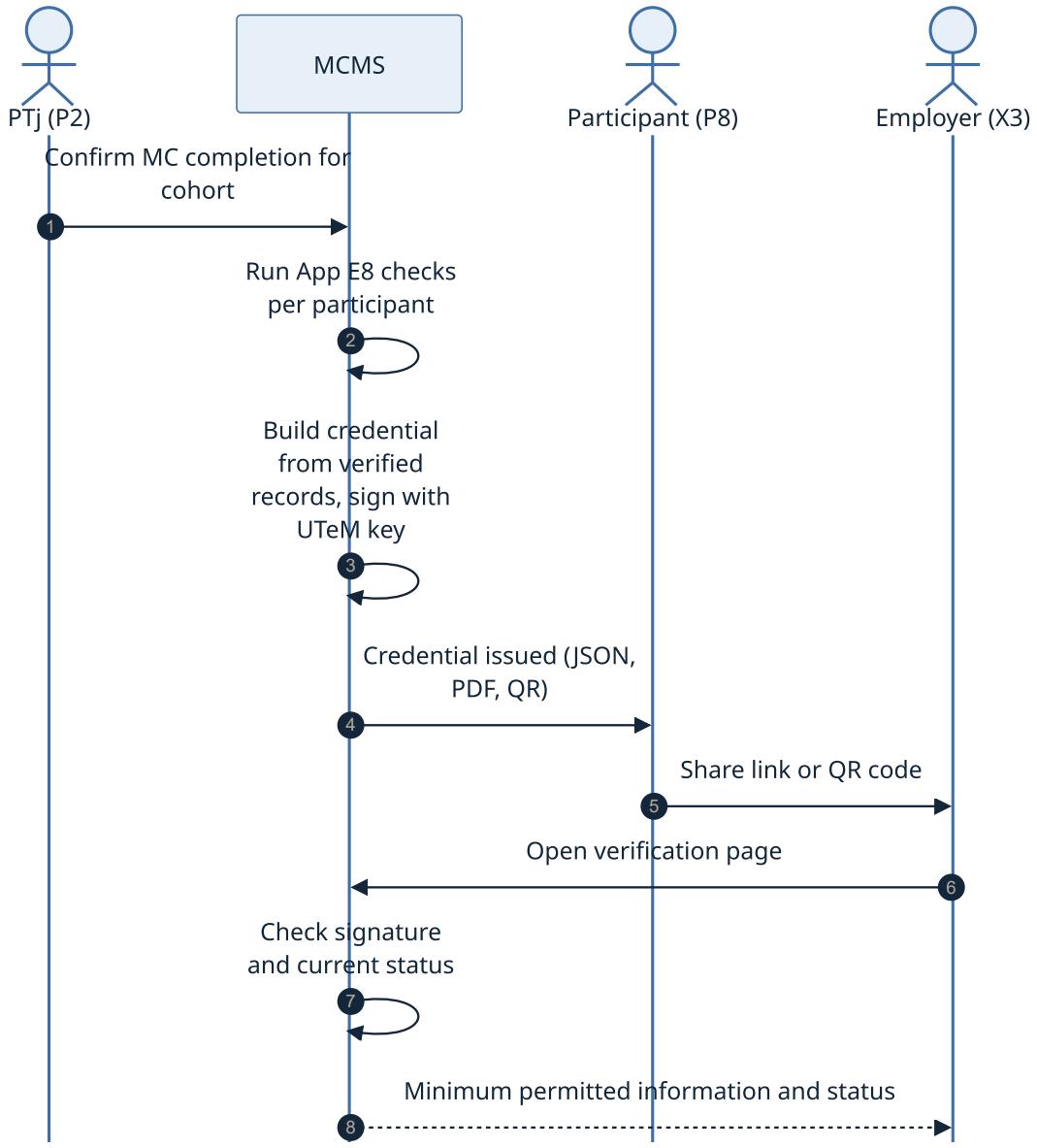


Figure 24: *Issuing a credential and verifying it publicly.*

A.5 End-to-End Participant Journey

The diagram follows one participant from a modular UMC to a stackable degree, showing every separate decision on the way.

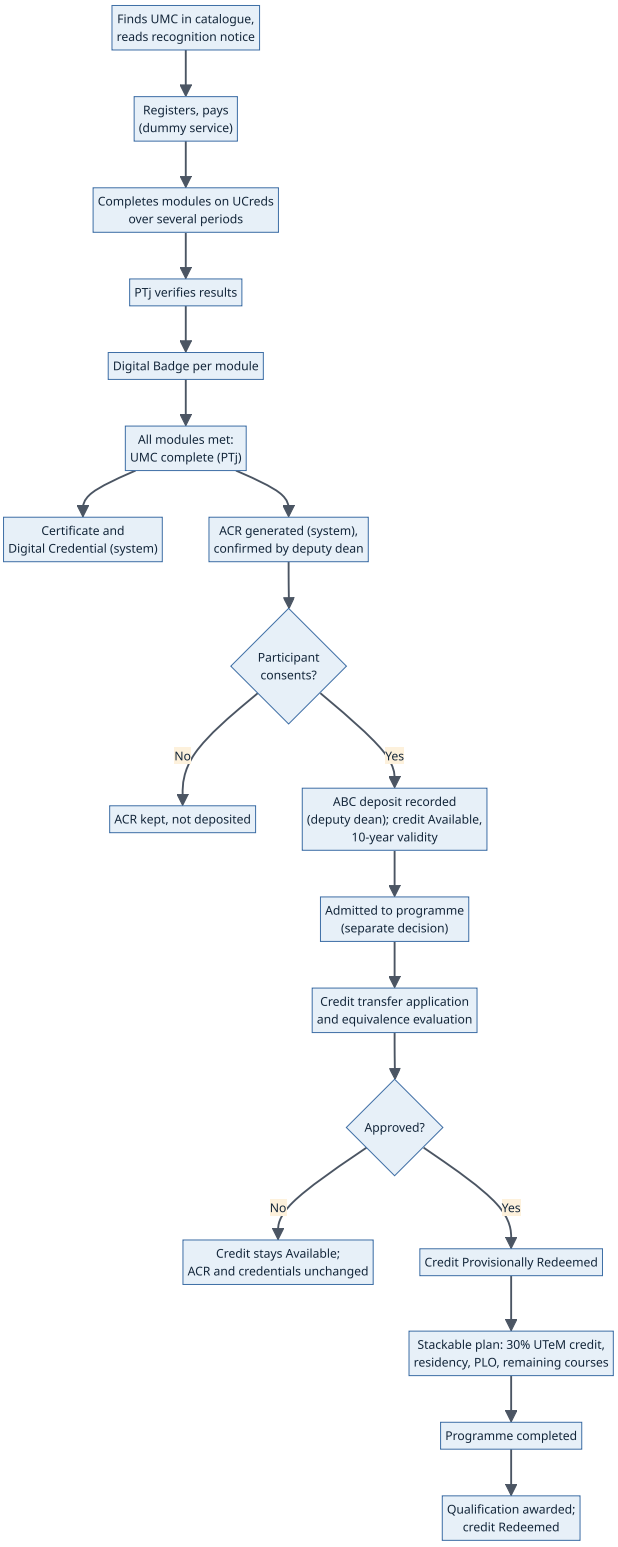


Figure 25: *End-to-end journey of a participant through MCMS.*

Appendix B: Traceability to the Guide

The table maps each part of the Guide that affects the system to the sections of this SRS that implement it. It lets the supervisor check coverage and lets developers find the source of a rule.

Table 14: *Traceability from Guide chapters and appendices to SRS sections*

Guide part	Subject	SRS sections	Business rules
Bab 1–5	Introduction, purpose, scope, definitions	1, Glossary	–
Bab 6	Concept and categories (UMC, SAMC, MCE), category determination	2.4, 3.2.1, 3.2.3	BR-01, BR-07, BR-08, BR-09, BR-10, BR-16, BR-17
Bab 7	Principles: outcomes, competence, flexibility, QA, improvement	3.2.2, 3.7, 3.15	BR-21
Bab 8	Framework: outcomes, SLT, content, assessment, certification, records	3.2.2, 3.7, 3.8, 3.9	BR-13, BR-14, BR-23, BR-27
Bab 9	Development, review, approval, offering	3.2, 3.3, 3.4	BR-19, BR-20
Bab 10	Delivery, platform, participants, assessment, certification, records	3.4.2, 3.5, 3.6, 3.7, 3.9	BR-22, BR-24
Bab 11	Quality assurance, Quality Verification, monitoring, review	3.10, 3.15	BR-02, BR-03, BR-25, BR-18
Bab 12	Credit accumulation, ACR, ABC, validity, single redemption	3.11, 3.13	BR-04, BR-26, BR-28, BR-29, BR-30, BR-31
Bab 13	Credit transfer	3.12	BR-32, BR-33, BR-34, BR-35, BR-15
Bab 14	Stackable degree	3.14	BR-05, BR-06, BR-36, BR-37, BR-38, BR-39
Bab 15	Governance structure, separation of decisions, traceability	2.5, 3.1, 3.3, 3.16	BR-18, BR-44
Bab 16	Roles and responsibilities	2.5, 3.1.2	BR-44
Bab 17	Monitoring and CQI, changes, versions, suspension	3.3.3, 3.15	BR-20, BR-43, BR-42
Bab 18	Records and documentation	3.8, 3.16, 6	BR-40, BR-41, BR-42
Lampiran A	Development and approval template	3.2.2, 3.2.3, 3.3	–
Lampiran B	Readiness and participant records checklist	3.4.1, 3.5	–
Lampiran C	QA, monitoring and CQI template	3.15	–
Lampiran D	QV evidence list for SAMC	3.10	–
Lampiran E	Badge, certificate and Digital Credential specification	3.9	–
Lampiran F	ACR, ABC, accumulation and transfer records	3.11, 3.12, 3.13	–
Lampiran G	Stackable degree mapping matrix	3.14	–
Lampiran H	Governance, RACI and records matrix	2.5, 3.1, 3.16, 6.3	–

Appendix C: Issues List

Open items OI-01 to OI-08 come from the Glossary and Decision Log v0.2. Items OI-09 to OI-15 were found while writing this SRS and have been entered in the Glossary and Decision Log v0.3. The Glossary and Decision Log v0.6 (MCMS_FND_V0.6_Oct_2026) is the controlling list; in it OI-01, OI-05, OI-06 and OI-14 are closed.

Table 15: *Open items*

ID	Question	Needed by	Working assumption	Affects
OI-01	Which Guide version is controlling: V2.0 (15 Sep 2026) or V1.0 (18 Sep 2026)?	Closed	The project's copy, Versi 1.0, 18 Sep 2026 (D-03)	Source tags throughout
OI-02	Is a PostgreSQL server available, or must the system use MySQL/MariaDB?	Before SDD	PostgreSQL, portable queries (D-09)	2.6, NFR-30
OI-03	Which SSO protocol and tenant settings does UTeM allow? How are outside applicants recognised?	Before SDD	OpenID Connect; account attribute or imported list (D-10)	FR-01, FR-04, IR-19
OI-04	How does UCreds connect (web service, data export, Moodle link) and does it issue badges?	Next iteration	CSV import; UCreds issues badges (D-13)	3.6, FR-117
OI-05	Is a separately credit-bearing UMC module an approved case at UTeM, and who approves it?	Closed	Case-by-case academic approval; approving role is a setting (D-18)	FR-60, FR-208
OI-06	Which committee or role approves each decision type?	Closed	Deputy dean endorses, Senate approves (recorded by the deputy dean), degree-owning deputy dean decides transfers (D-11)	FR-17
OI-07	Retention periods for each record category.	Before go-live	Placeholder settings; ACR indefinite [G 12.8]	FR-265, Table 12
OI-08	Who owns the credential signing keys and the public verification page?	Before SDD	CODL (D-12)	SR-08, IR-14
OI-09	What are the credit transfer limits and non-transferable components of each target programme, and who maintains them?	Before Release 1 testing	Programme settings entered by the faculty; warning only	FR-207
OI-10	How will MCMS learn of programme admission, completion and withdrawal, which drive redemption status?	Before SDD	Manual recording by an authorised user	FR-222, FR-239
OI-11	What data and file format does ABC require for deposit, and how are redemption updates sent to ABC?	Before Release 1 testing	Export of ACR fields of Appendix F3 in CSV and PDF	FR-191
OI-12	What fee rules, refund rules and sponsorship arrangements apply?	Before Release 2	Fee per offering; refunds outside MCMS	FR-103, FR-104
OI-13	What is the completion window for a modular UMC (Guide 7.4 "tempoh kesahan pembelajaran")?	Before Release 1 testing	Set per offering; no default	FR-135
OI-14	Confirm the conversion of 40 notional learning hours to 1 credit for UMC and SAMC calculations.	Closed	Confirmed: 40 hours = 1 credit (D-27)	Table 7

ID	Question	Needed by	Working assumption	Affects
OI-15	Which identity verification methods are required for online assessment at each risk level?	Before Release 1 testing	Recorded per assessment; method chosen by PTj	FR-124

Appendix D: Document History

Version	Date	Author	Change
0.1	23 October 2026	Aliff, Intern	Informal draft (sections 1 to 2.8); input material only.
1.0	23 October 2026	Aliff, Intern	First formal SRS, written against Glossary and Decision Log v0.2, in the layout of the reference SRS (D-04, D-26). Adds functional requirements for all sixteen modules, business rules, interface, data and non-functional requirements, diagrams, traceability and the issues list.
1.0 (rev. 1)	26 October 2026	Aliff, Intern	Aligned with Glossary and Decision Log v0.4: the Guide copy is fixed (D-03), module credit approval is academic and case by case (D-18), 1 credit = 40 hours (D-27); OI-01, OI-05 and OI-14 closed.
1.0 (rev. 2)	26 October 2026	Aliff, Intern	Aligned with Glossary and Decision Log v0.5: users simplified to Student, Lecturer, Deputy dean, Senate (no login), CODL (administration only) and public verifiers (D-28); credentials and ACR produced by the system after deputy dean verification (D-29); RACI, diagrams and requirements updated accordingly.
1.1	27 October 2026	Aliff, Intern	Aligned with Glossary and Decision Log v0.6 after the cross-document review of 27 October 2026: document set now includes the UI/UX Design Specification and the System Flow Map (D-30); Release 1 prototype defined (D-31); corporate identity assumption recorded (D-33, A-10); completion window (FR-133, OI-13) may be set per offering and sends no reminder while unset. No requirement ID changed.
1.2	28 October 2026	Aliff, Intern	Updated after the Release 1 prototype was built and its four-act scenario passed: the settings table lists the settings the prototype added (quality targets, periodic review, the registered-provider flag, the retention periods of the logs, the demo clock) and the approval routes for transfer, stackable mapping, suspension and disposal. No requirement ID or wording changed. Findings of the build that affect the SRS are in the Release 1 review of 28 October 2026 (for example: the module field of an assessment component, and the repeated import file, which is refused).
1.3	6 November 2026	Aliff, Intern	Aligned with Glossary and Decision Log v0.9 (D-39 to D-44): updated authentication guidance (fr-signinguide), credential verification URLs with Crockford Base32 and dynamic host resolution (fr-crid), verification result slip certificate layout with statutory PDPA notice and direct cryptographic file downloads (fr-vffields), machine JWKS discovery separation (fr-vftrust), UI badge simplification vs operational status indicators (ir-badges), articulation modal dialog (fr-spdefine), and institutional CODL branding (ir-identity).

Review of this document: under D-23 the PRD, SDD and Release 1 prototype proceed on the recorded assumptions; the supervisor's comments will be applied to all documents together once the prototype can be shown.